UNIT - I
Methods of Design of Concrete Structures

Design of Reinforced Concrete Structures (23A01502) - handwritten study notes

1. Methods of Design - Overview

- **Reinforced concrete (RC)** = concrete (strong in compression) + steel (strong in tension), acting together because of **bond**, equal thermal expansion and cover protection.
- Aim of structural design: the structure must be **safe** (strength, stability), **serviceable** (deflection, cracking, vibration), **durable** and **economical**.
- Code used: **IS 456:2000** (Plain and Reinforced Concrete - Code of Practice). Loads: IS 875, seismic: IS 1893. Design aids: SP 16.

Elastic method (WSM) → Ultimate load method (ULM) → Limit state method (LSM)

Method	Basis	Safety provided by
Elastic / Working stress	Linear stress-strain, working loads	Permissible stresses (factor of safety on materials)
Ultimate load	Collapse load, non-linear stress block	Load factor on working loads
Limit state	Probability, limit states	Partial safety factors on loads AND materials

Concept of Elastic Method (Working Stress Method)

- Materials behave **elastically**; stress is proportional to strain up to working loads.
- Stresses under **working (service) loads** are kept below **permissible stresses** = (strength / factor of safety).

- Concrete: permissible stress about 1/3 of cube strength; steel: about 0.55 fy.
- Uses **modular ratio m** to convert steel to an equivalent area of concrete (transformed section).
- Drawback: ignores non-linear behaviour, creep, shrinkage; no idea of real margin against failure.

Concept of Ultimate Load Method (ULM)

- Also called **load factor method**. Working loads are multiplied by a **load factor** (1.5 - 2.0) to get ultimate loads, and the section is designed to just fail at that load.
- Uses the **actual non-linear** stress-strain behaviour of concrete at failure (rectangular / parabolic stress block).
- Ultimate moment (rectangular block): $M_u = A_{st} f_y (d - A_{st} f_y / (1.7 f_{ck} b))$
- Advantages: more realistic, economical sections, rational. Drawback: **serviceability (deflection, cracks) not checked** - slender sections.

Concept of Limit State Method (LSM)

- A **limit state** is the condition beyond which the structure ceases to fulfil its intended function.
- LSM considers **both** safety (collapse) and serviceability together, using **partial safety factors** for loads and materials. It is the **main method of IS 456:2000**; WSM is retained as an alternative (Annex B).

Summary: WSM ⟶ stresses at working load. ULM ⟶ strength at ultimate load. LSM ⟶ collapse + serviceability with partial safety factors (best of both).

2. Working Stress Method as per IS 456 (Annex B)

Assumptions (B-1.3)

- At any cross-section, **plane sections remain plane** after bending.
- All tensile stresses are taken by **steel only** - concrete in tension is neglected.
- Stress is proportional to strain in concrete (**linear elastic**) and in steel.
- The **modular ratio** $m = E_s / E_c$ is the same for steel in tension and compression (equivalent area of steel = $m\ A_s$).
- Perfect **bond** between steel and concrete.

Modular Ratio and Material Constants

$$m = 280 / (3\ \sigma_{cbc})$$

- σ_{cbc} = permissible compressive stress in concrete in bending. (Here $E_c = 5000\ \sqrt{f_{ck}}$, $E_s = 2 \times 10^5\ N/mm^2$.)
- Permissible stresses (IS 456 Tables 21, 22 - flexure):

Grade	σ_{cbc} (bending)	σ_{cc} (direct)	$m = 280/(3\sigma_{cbc})$
M20	7.0	5.0	13.33
M25	8.5	6.0	10.98
M30	10.0	8.0	9.33
M35	11.5	9.0	8.12
M40	13.0	10.0	7.18
Steel	Fe 250	Fe 415	Fe 500
σ_{st} (N/mm^2)	140 (up to 20 mm dia), 130 (above)	230	275

Permissible shear stress and bond stress are also taken from IS 456 Tables 23 and 21 (bond stress for plain bars in M20 = 0.8 N/mm^2; increased by 60% for deformed bars).

Behaviour of a Singly Reinforced Section (WSM)

- Let b = width, d = effective depth, x = depth of neutral axis (NA), A_{st} = tension steel.
- **NA is located** by taking moment of areas about the NA: $b\,x^2/2 = m\,A_{st}\,(d - x)$ for the cracked section.
- Critical (balanced) NA: concrete and steel reach permissible stresses **simultaneously**.

$$k = x_c/d = m\,\sigma_{cbc} / (m\,\sigma_{cbc} + \sigma_{st})$$

$$j = 1 - k/3 \text{ (lever arm } z = j\,d)$$

- Moment of resistance (concrete side): $M_c = (1/2)\,\sigma_{cbc}\,k\,j\,b\,d^2 = Q\,b\,d^2$, where $Q = 0.5\,\sigma_{cbc}\,k\,j$.
- Moment of resistance (steel side): $M_s = \sigma_{st}\,A_{st}\,j\,d$.
- Percentage of steel for balanced section: $p_{t,bal} = 50\,k\,\sigma_{cbc} / \sigma_{st}$ (%).

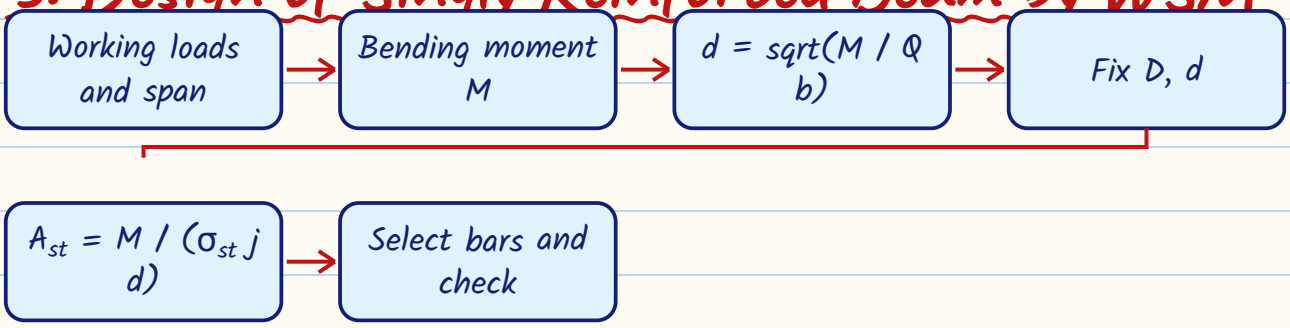
M20 with	k	j	Q (N/mm^2)	$p_{t,bal}$ %
Fe 250 (σ_{st}=140)	0.400	0.867	1.21	1.00
Fe 415 (σ_{st}=230)	0.289	0.904	0.913	0.439
Fe 500 (σ_{st}=275)	0.253	0.916	0.812	0.323

Types of Section (WSM)

- **Balanced** (critical): $x = x_c$. Steel and concrete reach limits together; $A_{st} = A_{st,bal}$.
- **Under-reinforced**: A_{st} < balanced, $x < x_c$. **Steel yields first** ⟶ ductile, warning before failure; $M = \sigma_{st}\,A_{st}\,j\,d$.
- **Over-reinforced**: A_{st} > balanced, $x > x_c$. **Concrete crushes first** (brittle, sudden). In WSM, $M = (1/2)\,\sigma_{cbc}\,b\,x\,(d - x/3)$ governed by concrete.

IS 456 (WSM): the section may be under- or over-reinforced; under-reinforced preferred, because of ductile failure and warning. In LSM, over-reinforced sections are NOT permitted.

3. Design of Singly Reinforced Beam by WSM



Design Steps

- 1. Assume b (about D/2 to D/3) and grade of concrete and steel; get σ_{cbc}, σ_{st}, m.
- 2. Compute k, j, Q for the balanced section.
- 3. Working moment $M = w\,l^2 / 8$ (simply supported beam, UDL including self weight).
- 4. Required effective depth $d = \sqrt{M / (Q\,b)}$; adopt a **greater** round value; D = d + cover + half bar dia.
- 5. $A_{st} = M / (\sigma_{st}\,j\,d)$; provide bars and check $A_{st,min}$, spacing, and bond.
- 6. Check shear and deflection (span/depth). Provide stirrups.

Worked Example 1 (WSM)

Q. Design a simply supported RC beam of effective span 5 m carrying a total working load of 15 kN/m (including self weight). Use M20 and Fe 415. Take b = 250 mm.

- $\sigma_{cbc} = 7$, $\sigma_{st} = 230$, $m = 13.33$. $k = 93.33/(93.33+230) = 0.289$, $j = 1 - 0.289/3 = 0.904$, $Q = 0.913$.
- $M = w\,l^2/8 = 15 \times 5^2 / 8 =$ **46.875 kN.m**
- $d = \sqrt{46.875 \times 10^6 / (0.913 \times 250)} = \sqrt{205{,}367} =$ **453 mm**.

Adopt d = 455 mm, overall depth D = 500 mm (effective cover 45 mm).

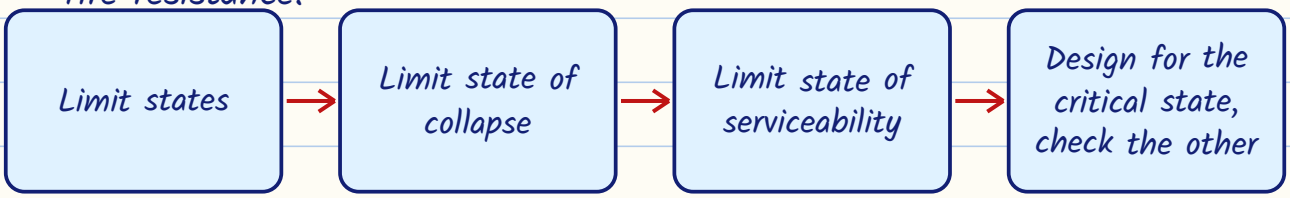
- A_{st} = 46.875 x 10^6 / (230 x 0.904 x 455) = **496 mm^2**. Provide 3 - 16 mm dia bars (603 mm^2).
- Check actual NA: 125 x^2 = 13.33 x 603 (455 - x) ⟶ x = 141.9 mm > x_c = 0.289 x 455 = 131.4 mm. The section is slightly over-reinforced, concrete governs.
- Actual M = 0.5 x 7 x 141.9 x 250 x (455 - 141.9/3) = **50.6 kN.m > 46.9 kN.m** ⟶ safe.

Final: b x D = 250 x 500, 3 - 16φ at bottom, 2 - 12φ hanger bars at top, 8 mm 2-legged stirrups (spacing from shear design).

4. Limit State Philosophy as per IS 456

Limit States

- **Limit state of collapse (strength)**: flexure, compression, shear, torsion; overturning, sliding, buckling, rupture.
- **Limit state of serviceability**: **deflection**, **cracking**, durability, vibration, fire resistance.



Characteristic Loads and Strengths

- **Characteristic strength of concrete f_{ck}** = 28-day cube strength below which not more than **5%** of test results fall.
- **Characteristic strength of steel f_y** = yield (0.2% proof) stress with 5% defective probability.
- **Characteristic load** = load with not more than 5% probability of being exceeded during the design life.

- **Design value** = characteristic value / partial safety factor (for materials) or characteristic value x factor (for loads).

Partial Safety Factors

Material	γ_m	Design strength
Concrete	1.5	$f_d = 0.446\ f_{ck}$ (= $0.67\ f_{ck}/1.5$)
Steel	1.15	$f_{sd} = 0.87\ f_y$ (= $f_y/1.15$)

Partial Safety Factors for Loads (IS 456 Table 18)

Load combination	DL	IL	WL / EL
DL + IL	1.5	1.5	-
DL + WL (or EL)	1.5 or 0.9*	-	1.5
DL + IL + WL (or EL)	1.2	1.2	1.2

- *0.9 DL is used when stability against overturning or stress reversal is critical.
- Design load (**factored load**) = γ_f x characteristic load. Eg. w_u = 1.5 (DL + LL).

Assumptions in Flexure (IS 456 Cl. 38.1)

- Plane sections remain plane (strain varies linearly).
- **Maximum strain in concrete** at the extreme compression fibre = **0.0035** in bending.
- Stress-strain curve of concrete: **parabolic up to strain 0.002, then constant** at 0.446 f_{ck} (design).
- **Tensile strength of concrete is ignored.**
- Stress in steel from the design stress-strain curve (Fe 250: 0.87 f_y after yield; Fe 415/500: non-linear curve).
- Max strain in tension steel at failure must be at least **$(0.87\ f_y/E_s)$ + 0.002** (ductility), which restricts x_u to $x_{u,max}$.

Equivalent Rectangular Stress Block

- Parabolic-rectangular area is replaced by an equivalent rectangle:

Compression $C = 0.36\ f_{ck}\ b\ x_u$

Centroid of C from top = $0.42\ x_u$ (lever arm $z = d - 0.42\ x_u$)

Tension $T = 0.87\ f_y\ A_{st}$

5. Advantages of Limit State Method over Other Methods

WSM	LSM
Linear elastic assumption not true near failure	Considers real non-linear behaviour
Single factor of safety on materials only	Separate partial factors on loads and materials
Factor of safety against collapse is not known	Reliable margin against collapse is known
Serviceability not considered directly	Checks both strength and serviceability
Ignores creep, shrinkage, redistribution	Allows moment redistribution and better detailing
Uneconomical (heavier sections)	Economical, about 10-15% saving of steel
Same safety for DL and LL	Different factors reflect uncertainty of each load

6. Analysis of Singly Reinforced Rectangular Beam by LSM

C = T equilibrium → Find x_u → Compare with $x_{u,max}$ → Find M_u = T x lever arm

Neutral Axis Depth

$0.36\ f_{ck}\ b\ x_u = 0.87\ f_y\ A_{st} \longrightarrow x_u = 0.87\ f_y\ A_{st} / (0.36\ f_{ck}\ b)$

Limiting Depth of Neutral Axis (IS 456 Cl. 38.1, Table in Annex G)

- At the **limiting** (balanced) condition, concrete strain = 0.0035 and steel strain = $0.87 f_y/E_s + 0.002$.

$$x_{u,max}/d = 0.0035 / (0.0055 + 0.87 f_y/E_s)$$

Steel	$x_{u,max}/d$	$M_{u,lim}/(f_{ck} b d^2)$	$p_{t,lim}$ (M20) %
Fe 250	0.531	0.148	1.76
Fe 415	0.479	0.138	0.96
Fe 500	0.456	0.133	0.76
Fe 550	0.440	0.129	0.66

Types of Section in LSM

- **Under-reinforced**: $x_u < x_{u,max}$ (steel yields first; ductile). Preferred. $M_u = 0.87 f_y A_{st} (d - 0.42 x_u)$.
- **Balanced**: $x_u = x_{u,max}$. $M_{u,lim} = 0.36 f_{ck} b x_{u,max} (d - 0.42 x_{u,max})$.
- **Over-reinforced**: $x_u > x_{u,max}$. **Not permitted**; take $M_u = M_{u,lim}$ (steel does not yield, brittle failure).

Moment of Resistance

$$M_u = 0.87 f_y A_{st} d (1 - A_{st} f_y / (b d f_{ck}))$$

$$M_{u,lim} = 0.138 f_{ck} b d^2 \text{ (Fe 415) } 0.148 \text{ (Fe 250) } 0.133 \text{ (Fe 500)}$$

- Using percentage of steel $p_t = 100 A_{st}/(b d)$:

$$M_u / (b d^2) = 0.87 f_y (p_t/100) (1 - f_y p_t / (100 f_{ck}))$$

Design Formula for A_{st} (when M_{u} <= M_{u,lim})

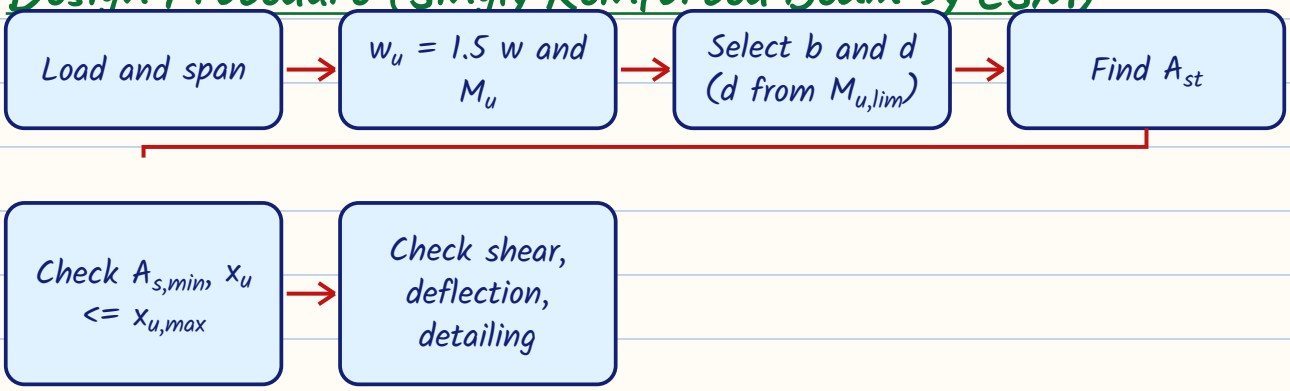
$$A_{st} = (0.5 f_{ck} / f_y) [1 - \text{sqrt}(1 - 4.6 M_u / (f_{ck} b d^2))] b d$$

Limits on Steel (IS 456 Cl. 26.5.1)

- **Minimum tension steel**: $A_{s,min} = 0.85 b d / f_y$ (Fe 415: 0.205% of bd).

- **Maximum tension steel**: 0.04 b D (4% of gross area).
- Side face steel 0.1% of web area if D > 750 mm (spaced not more than 300 mm).

Design Procedure (Singly Reinforced Beam by LSM)



Worked Example 2 (SRS beam by LSM)

Q. A simply supported beam of effective span 6 m carries imposed load 12 kN/m and dead load 8 kN/m (excluding self weight). Design for flexure. Use M20, Fe 415. Take b = 250 mm, D = 550 mm, clear cover 30 mm, using 22 mm bars and 8 mm stirrups.

- Effective depth d = 550 - 30 - 10 (approx half bar plus stirrup adjustment) = **510 mm**. Self weight = 0.25 x 0.55 x 25 = 3.44 kN/m.
- Total w = 8 + 12 + 3.44 = 23.44 kN/m ⟶ w_u = 1.5 x 23.44 = **35.16 kN/m**.
- $M_u = w_u l^2/8 = 35.16 \times 6^2/8$ = **158.2 kN.m**
- $M_{u,lim} = 0.138 \times 20 \times 250 \times 510^2$ = **179.5 kN.m** > 158.2 kN.m ⟶ singly reinforced section is adequate.
- $M_u/(b\,d^2) = 158.2 \times 10^6 / (250 \times 510^2) = 2.433$ N/mm²; 4.6 x 2.433 / 20 = 0.5596 ⟶ sqrt(1 - 0.5596) = 0.6636
- $A_{st} = (0.5 \times 20 / 415) \times (1 - 0.6636) \times 250 \times 510$ = **1033 mm²**
- Provide **3 - 22φ (1140 mm²)**. p_t = 1140/(250 x 510) = 0.894%. $A_{s,min}$ = 0.85 x 250 x 510 / 415 = 261 mm² <1140 OK, < 4% bD.
- Check $x_u = 0.87 \times 415 \times 1140 / (0.36 \times 20 \times 250)$ = **228.7 mm** < $x_{u,max}$ = 0.48 x 510 = **244.8 mm** ⟶ under-reinforced OK.

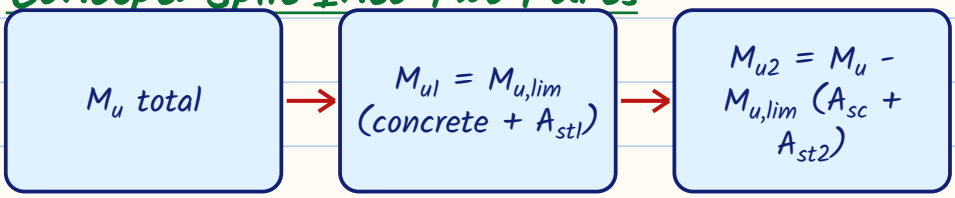
- Actual M_u = 0.87 x 415 x 1140 x (510 - 0.42 x 228.7) = **170.4 kN.m** > **158.2** ⟶ safe.
- Spacing: (250 - 2 x 30 - 2 x 8 - 3 x 22)/2 = 54 mm > 25 mm and > bar dia OK. Span/depth = 6000/510 = 11.8 < 20 OK.

Final: 250 x 550, 3 - 22φ at bottom, 2 - 12φ at top, 8φ stirrups.

7. Doubly Reinforced Rectangular Beams

- Used when **M_u > $M_{u,lim}$** and section size can not be increased (architectural or head-room limit), or when moment reversal occurs.
- Steel is also provided in the **compression zone** (A_{sc}) in addition to tension steel (A_{st}).
- Advantages: increases moment capacity, **ductility**, reduces long-term deflection (compression steel resists creep), helps stirrup support.

Concept: Split Into Two Parts



- x_u is taken equal to **$x_{u,max}$** for the design case (so steel yields).

$$M_{u1} = M_{u,lim} = 0.36\, f_{ck}\, b\, x_{u,max}\, (d - 0.42\, x_{u,max})$$

$$M_{u2} = M_u - M_{u,lim} = A_{sc}\, (f_{sc} - f_{cc})\, (d - d')$$

- f_{cc} = compressive stress in concrete at the level of compression steel = **0.446 f_{ck}** (to avoid double counting of the concrete area).

$$A_{sc} = M_{u2} / [\, (f_{sc} - 0.446\, f_{ck})\, (d - d')\,]$$

$$A_{st1} = 0.36\, f_{ck}\, b\, x_{u,max} / (0.87\, f_y)$$

$$A_{st2} = A_{sc} (f_{sc} - f_{cc}) / (0.87 f_y) \quad A_{st} = A_{st1} + A_{st2}$$

Stress in Compression Steel (fsc)

- Strain at the level of A_{sc}: $\varepsilon_{sc} = 0.0035 (1 - d'/x_{u,max})$; f_{sc} is read from the stress-strain curve (SP 16 Table).

d'/d	0.05	0.10	0.15	0.20
f_{sc} Fe 250	217	217	217	217
f_{sc} Fe 415	355	353	342	329
f_{sc} Fe 500	424	412	395	370

- For Fe 250: $f_{sc} = 0.87 f_y = 217.5$ N/mm^2. d' = distance from the compression face to centroid of A_{sc} (effective cover).

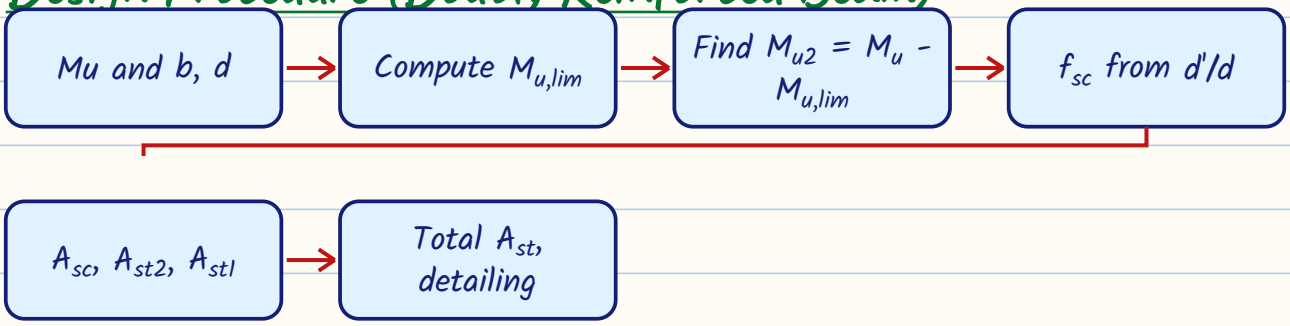
Analysis of a Given Doubly Reinforced Section

- Assume x_u; find ε_{sc}, f_{sc}; solve equilibrium C = T:

$$0.36 f_{ck} b x_u + A_{sc} (f_{sc} - f_{cc}) = 0.87 f_y A_{st}$$

- If $x_u <= x_{u,max}$: $M_u = 0.36 f_{ck} b x_u (d - 0.42 x_u) + A_{sc} (f_{sc} - f_{cc}) (d - d')$.
- If $x_u > x_{u,max}$: take $x_u = x_{u,max}$ (limit) and compute the same expression.

Design Procedure (Doubly Reinforced Beam)



- Provide **stirrups** (closely spaced) around compression bars to prevent buckling: spacing <= 16 x dia of compression bar, <= 300 mm.

Worked Example 3 (Doubly Reinforced Beam)

Q. A beam 250 mm x 500 mm (effective cover 50 mm both sides) has

factored moment M_u = 230 kN.m. Design the reinforcement. Use M20 and Fe 415.

- d = 500 − 50 = **450 mm**, d' = 50 mm, d'/d = 0.111.
- $M_{u,lim}$ = 0.138 × 20 × 250 × 450^2 = **139.7 kN.m** < 230 ⟶ **doubly reinforced** section needed.
- M_{u2} = 230 − 139.7 = **90.3 kN.m**
- $x_{u,max}$ = 0.48 × 450 = 216 mm. ε_{sc} = 0.0035 (1 − 50/216) = 0.00269 ⟶ f_{sc} = **351 N/mm²** (from table, by interpolation of d'/d = 0.111).
- f_{cc} = 0.446 × 20 = 8.92 N/mm²
- A_{sc} = 90.3 × 10^6 / ((351 − 8.92)(450 − 50)) = 90.3 × 10^6 / 136,832 = **660 mm²** ⟶ 3 − 18φ (763 mm²).
- A_{st1} = 0.36 × 20 × 250 × 216 / (0.87 × 415) = 388,800 / 361.05 = **1077 mm²**
- A_{st2} = A_{sc} (f_{sc} − f_{cc})/(0.87 f_y) = 660 × 342.08 / 361.05 = **625 mm²**
- A_{st} = 1077 + 625 = **1702 mm²** ⟶ provide 6 − 20φ (1885 mm²) in two layers (3 + 3).
- Check max steel: 4% of bD = 0.04 × 250 × 500 = 5000 mm² > 1885 OK; A_{sc} = 763 < 5000 OK.

Result: tension steel 6 − 20φ (two layers), compression steel 3 − 18φ, stirrups 8φ at close spacing. Always check d'/d and $x_{u,max}$.

8. Detailing Requirements (IS 456 Cl. 26)

Cover to Reinforcement (Table 16, beams/slabs)

Exposure	Nominal cover (mm)
Mild	20
Moderate	30
Severe	45
Very severe	50

Extreme	75

- Cover must be at least bar dia and ensures **fire resistance, corrosion protection, bond**.

Spacing of Bars

- Minimum horizontal clear spacing: **the larger of** bar diameter, **25 mm**, or (maximum aggregate size + 5 mm).
- Vertical spacing between layers: >= 15 mm, 2/3 of maximum aggregate size, or bar dia.
- **Maximum clear spacing** in tension zone (Table 15): Fe 250: 300 mm; Fe 415: 180 mm; Fe 500: 150 mm (for no redistribution).

Effective Span, Depth and Control of Deflection

- **Effective span** of simply supported beam = lesser of (clear span + d) and (centre to centre of supports).
- **Effective depth d** = D - effective cover (cover + stirrup + half bar dia).
- **Basic span/depth ratios** (Cl. 23.2.1): cantilever **7**, simply supported **20**, continuous **26** (for span up to 10 m). Spans > 10 m: multiply by 10/span in metres (not for cantilevers).
- Modification factors: for **tension steel** (Fig. 4: depends on $f_s = 0.58\ f_y\ A_{st,req}/A_{st,prov}$ and p_t), **compression steel** (Fig. 5), **flanged beams** (Fig. 6).

$$(l/d)_{max} = (\text{basic } l/d) \times k_t \times k_c \times k_f$$

- Anchorage of tension bars: $L_d = \varphi\ \sigma_s / (4\ \tau_{bd})$; continued past the support by L_d.

Remember (beam): Ast,min = 0.85 bd/fy ; Ast,max = 0.04 bD ; cover 20-75 ; l/d = 7, 20, 26.

9. Quick Revision

- WSM: elastic, permissible stress, $m = 280/(3\,\sigma cbc)$, $M = Q\,b\,d^2$, $Ast = M/(\sigma st\,j\,d)$.
- ULM: load factor 1.5 – 2, non-linear block; no serviceability check.
- LSM: $\gamma c = 1.5$, $\gamma s = 1.15$, loads 1.5 / 1.2; 0.446 fck, 0.87 fy; $C = 0.36\,fck\,b\,xu$; $z = d - 0.42\,xu$.
- xu,max/d: **0.53** / **0.48** / **0.46** / **0.44** for Fe 250 / 415 / 500 / 550.
- Mu,lim = 0.148 / 0.138 / 0.133 / 0.129 x fck b d^2.
- DRS: $Mu2 = Asc\,(fsc - fcc)(d - d')$, $fcc = 0.446\,fck$, $Ast = Ast1 + Ast2$.

Important Questions

- 1. Explain the concepts of working stress, ultimate load and limit state methods. Compare them.
- 2. List the assumptions of WSM. Derive k, j and Q for a balanced singly reinforced section.
- 3. Define balanced, under-reinforced and over-reinforced sections. Which is preferred and why?
- 4. Design a singly reinforced beam by WSM for given span and load (M20, Fe 415).
- 5. Explain limit state philosophy. State characteristic strength, load and partial safety factors.
- 6. Derive the stress block parameters 0.36 fck b xu and 0.42 xu. Derive $x_{u,max}/d$ for Fe 415.
- 7. Derive $M_{u,lim}$ for a singly reinforced rectangular beam. Why is the over-reinforced section not permitted?
- 8. Write the advantages of LSM over WSM.
- 9. Design a doubly reinforced beam for the given M_u. Explain how f_{sc} is obtained.

UNIT - II
Flanged Beams, Bond, Shear & Torsion

Design of Reinforced Concrete Structures (23A01502) - handwritten study notes

1. Flanged Beams (T and L Beams)

Why flanged beams?

- In a monolithic RC floor, the **slab and beam cast together** act as one unit. A part of the slab on each side of the web works as the **flange** and takes compression.
- Cross-section is **T-shaped** (interior beam, slab both sides) or **L-shaped** (edge beam, slab on one side).
- Advantages: large compression area, so **less steel**, smaller depth, economical for long spans.

In a flanged beam the flange is in compression only in **sagging (span) zones**. Near supports (hogging) the flange is in tension, so the beam is designed as a **rectangular beam of width b_w**.

Slab (flange) b_f x D_f → Web b_w x (D - D_f) → Tension steel A_{st} at bottom

Effective flange width (IS 456 cl. 23.1.2)

Beam	Effective width b_f
T-beam (continuous or simply supported)	$l_o/6 + b_w + 6\,D_f$
L-beam (continuous or simply supported)	$l_o/12 + b_w + 3\,D_f$
Isolated T-beam	$l_o / (l_o/b + 4) + b_w$
Isolated L-beam	$0.5\,l_o / (l_o/b + 4) + b_w$

- l_o = distance between points of zero moment (simply supported span = effective span; continuous = about 0.7 x span).

- b_w = web width, D_f = flange (slab) thickness, b = actual width of flange.
- **Rule:** b_f must never exceed the actual centre-to-centre spacing of the beams.

Take the **least** of the calculated b_f and the actual spacing. Memorise: T $\longrightarrow l_o/6 + b_w + 6D_f$; L $\longrightarrow l_o/12 + b_w + 3D_f$.

Position of neutral axis

- Assume the NA at the **bottom of flange**, $x_u = D_f$, and find the moment of resistance of the flange:

$$M_{u,f} = 0.36\, f_{ck}\, b_f\, D_f\, (d - 0.42\, D_f)$$

- If $M_u <= M_{u,f}$: NA lies in the flange ($x_u <= D_f$). Design as a **rectangular beam of width b_f**.
- If $M_u > M_{u,f}$: NA lies in the web ($x_u > D_f$). Use the flanged-beam equations below.

Calculate b_f → Find $M_{u,f}$ ($x_u = D_f$) → Compare M_u with $M_{u,f}$ → Choose case 1 or case 2

Case 1: Neutral axis within flange ($x_u <= D_f$)

- Section behaves as a **rectangular section b_f x d**.

$$M_u = 0.87\, f_y\, A_{st}\, d\, (1 - A_{st}\, f_y / (b_f\, d\, f_{ck}))$$

- Depth of NA: $x_u = 0.87\, f_y\, A_{st} / (0.36\, f_{ck}\, b_f)$; check $x_u <= D_f$ and $x_u <= x_{u,max}$.
- Limiting values $x_{u,max}/d$: **0.53 (Fe250), 0.48 (Fe415), 0.46 (Fe500)**.

Case 2: Neutral axis in web ($x_u > D_f$)

Compression is taken by the web (rectangular stress block above NA) plus the **projecting flange**.

- **(a) When $D_f/d <= 0.2$** (thin flange, stress is nearly uniform at 0.447

f_{ck}):

$$M_u = 0.36\, f_{ck}\, b_w\, x_u\, (d - 0.42\, x_u) + 0.45\, f_{ck}\, (b_f - b_w)\, D_f\, (d - D_f/2)$$

$$0.87\, f_y\, A_{st} = 0.36\, f_{ck}\, b_w\, x_u + 0.45\, f_{ck}\, (b_f - b_w)\, D_f$$

- **(b) When $D_f/d > 0.2$** (thick flange, stress is not uniform), replace D_f by the equivalent depth y_f:

$$y_f = 0.15\, x_u + 0.65\, D_f \text{ (but } y_f <= D_f)$$

$$M_u = 0.36\, f_{ck}\, b_w\, x_u\, (d - 0.42\, x_u) + 0.45\, f_{ck}\, (b_f - b_w)\, y_f\, (d - y_f/2)$$

$$0.87\, f_y\, A_{st} = 0.36\, f_{ck}\, b_w\, x_u + 0.45\, f_{ck}\, (b_f - b_w)\, y_f$$

Procedure for $x_u > D_f$: assume x_u ⟶ find A_{st} from force equilibrium ⟶ compute M_u ⟶ adjust. Always check $x_u <= x_{u,max}$ (else beam is over-reinforced).

Design steps for a T-beam

Fix span, loads, fck, fy → Assume D = 1/12 to 1/15 and d, b_w → Find M_u = 1.5 M and b_f → Check $M_{u,f}$ (NA position) → Find A_{st}, verify x_u → Detail steel, check shear, deflection

- Ast limits: $A_{st,min} = 0.85\, b_w\, d / f_y$, $A_{st,max} = 0.04\, b_w\, D$.

Worked Example 1: Design of a T-beam

Given: simply supported T-beam, effective span 6 m, spacing of beams 3 m c/c, b_w = 300 mm, D_f = 100 mm, d = 500 mm, M25 and Fe415, factored moment M_u = 500 kN m. Find A_{st}.

- **Step 1 - flange width:** $b_f = l_o/6 + b_w + 6\,D_f = 6000/6 + 300 + 600 =$ **1900 mm** < 3000 mm spacing, so OK.
- **Step 2 - NA position:** $M_{u,f} = 0.36 \times 25 \times 1900 \times 100 \times (500 - 42) =$ **783 kN m**.
- Since $M_u = 500 < 783$: **NA lies in the flange**, so design as rectangular $b_f = 1900$, $d = 500$.
- **Step 3 - steel:** $500 \times 10^6 = 0.87 \times 415 \times A_{st} \times 500\,(1 - 415\,A_{st} / (1900 \times 500 \times 25))$.
- Simplify: $A_{st}^2 - 57229\,A_{st} + 1.585 \times 10^8 = 0$, so **$A_{st}$ = 2918 mm²**.
- **Step 4 - check:** $x_u = 0.87 \times 415 \times 2918 / (0.36 \times 25 \times 1900) =$ **61.6 mm < D_f** (OK) and < $x_{u,max} = 0.48 \times 500 = 240$ mm.
- $A_{st,min} = 0.85 \times 300 \times 500 / 415 = 307$ mm² < provided.

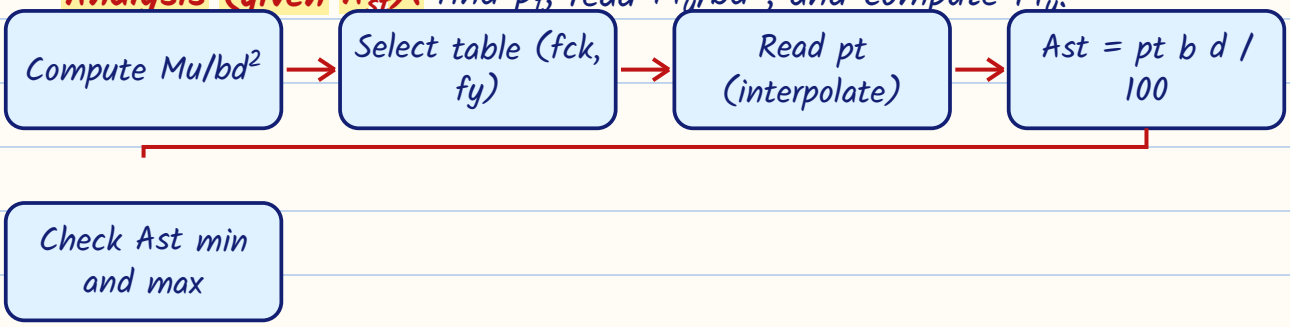
Provide 6 bars of 25 mm dia (A_{st} = 2945 mm²) in two layers, (3 + 3). Nominal cover 25 mm for mild exposure; add stirrups and check shear and deflection.

2. Use of Design Aids for Flexure (SP-16)

- **SP-16** (Design Aids for IS 456) gives tables and charts that avoid trial and error in flexure design.
- The tables give **$M_u/(b\,d^2)$** against **$p_t = 100\,A_{st}/(b\,d)$** for each grade of concrete.

SP-16 item	Content
Tables 1 to 4	Rectangular sections, singly reinforced: M_u/bd^2 vs p_t for fy 250, 415, 500
Tables 45 to 56	Doubly reinforced sections, with d'/d ratios
Tables 57 to 59	**Flanged beams**: $M_u/(b_w\,d^2)$ for varying b_f/b_w and D_f/d
Charts 1 to 18	Same data presented as curves

- **Procedure for design (given M_u, b, d):**
  - Compute $M_u / (b\ d^2)$ (in N/mm^2).
  - From the table for the given f_{ck} and f_y, read p_t (interpolate between rows).
  - $A_{st} = p_t\ b\ d / 100$.
- **Analysis (given A_{st}):** find p_t, read M_u/bd^2, and compute M_u.



Design aids give quick, reliable answers (accurate to about 1-2 %). For flanged beams find b_f, then check D_f/d and b_f/b_w to enter Tables 57-59.

3. Behaviour of RC Members in Bond and Anchorage

What is bond?

- **Bond** is the adhesion and grip between concrete and reinforcing steel which makes them act together (strain compatibility).
- Bond comes from three sources: **chemical adhesion**, **friction**, and **mechanical interlocking** of ribs (the most important for deformed bars).
- **Bond stress** tau_b = shear stress at the steel-concrete interface = change in bar force / (perimeter x length).
- Bond failure types: pull-out of the bar, splitting of the concrete cover (more common).

Design bond stress tau_{bd} (IS 456 cl. 26.2.1.1)

Grade of concrete	M20	M25	M30	M35	M40 and above
tau_{bd} for plain bars (N/mm^2)	1.2	1.4	1.5	1.7	1.9

Deformed bars (60 % more)	1.92	2.24	2.4	2.72	3.04

- For bars in **compression**, τ_{bd} is **increased by 25 %**.
- Deformed bars (HYSD, Fe415/500) give **60 % higher** τ_{bd} than plain bars.

Development length L_{d}

Length of bar required to develop its design stress ($0.87 f_y$) by bond.

$$L_d = \phi \sigma_s / (4 \tau_{bd}) = 0.87 f_y \phi / (4 \tau_{bd})$$

- ϕ = nominal diameter of bar, $\sigma_s = 0.87 f_y$ = stress in bar at section.
- **Fe415, M20, deformed:** $L_d = 0.87 \times 415 \phi / (4 \times 1.92) = 47 \phi$.
- Typical values for Fe415 deformed tension bars: M20 ⟶ 47 phi, M25 ⟶ 40 phi, M30 ⟶ 38 phi.

Use L_d for **anchoring** a bar beyond the section of maximum stress.
Hooks are not allowed for compression bars.

Anchorage by bends and hooks

Type	Anchorage value	Notes
Standard 180 deg hook (U-type)	16 phi	straight extension >= 4 phi (min 65 mm)
Standard 90 deg bend	8 phi	straight extension >= 4 phi
Min internal radius of bend	2 phi (mild steel), 4 phi (HYSD)	to avoid concrete crushing

Check for development length at supports (cl. 26.2.3.3)

For a **positive-moment (sagging) bar** at simple supports:

$$L_d <= 1.3 M_1 / V + L_o$$

- M_1 = moment of resistance of the section assuming all bars at the

section are stressed to 0.87 f_y.

- **V** = shear force at the section due to design loads; L_o = anchorage beyond the centre of support (equivalent anchorage of hook/bend). L_o is limited to **d or 12 phi, whichever is greater**.
- Factor **1.3** is used when the end of the reinforcement is confined by a compressive reaction (support).
- At the point of inflexion, **L_d <= M_1/V + L_o**.

This check controls the size of bars at simple supports: smaller diameters have smaller L_d, so choose closely spaced small bars.

Laps, splices and curtailment

- **Lap length in flexural tension >= L_d or 30 phi** (whichever is greater); compression >= L_d or 24 phi.
- Direct tension: lap >= 2 L_d or 30 phi. Do not lap bars larger than **36 mm** dia (use mechanical/welded splices).
- Laps should be staggered, away from points of maximum stress. Not more than **50 %** bars lapped at a section.
- **Curtailment:** each bar extends beyond its theoretical cut-off point by **d or 12 phi** (greater). At least **1/3** of positive steel reaches the support (simple support) and 1/4 for continuous.
- Never curtail in a tension zone unless shear is small (V_u <= 2/3 of shear resistance) or extra stirrups are provided.

Worked Example 2: Check for development length

Given: beam 300 x 550 (d = 500), 3 bars of 20 mm dia (A_{st} = 942 mm^2) continue to a simple support, M20 and Fe415, factored shear at support V = 180 kN. Check the development length.

- tau_{bd} = 1.2 x 1.6 = 1.92 N/mm^2 (deformed bars). L_d = 0.87 x 415 x 20 / (4 x 1.92) = **940 mm**.
- M_1 = 0.87 f_y A_{st} d (1 - A_{st} f_y/(b d f_{ck})) = 361 x 942 x 500 (1 - 0.13) = **148 kN m**.

- $M_1/V = 148 \times 10^6 / 180000 =$ **822 mm**. With support width 230 mm: $L_o = 115 - 25 = 90$ mm.
- Limit = $1.3 \times 822 + 90 =$ **1158 mm** > $L_d = 940$ mm. Hence **OK, safe in bond.**

If the check fails: use smaller diameter bars (more of them), or provide a standard 90 deg bend / hook, or increase support width.

4. Behaviour of RC Beams in Shear

Diagonal tension

- Shear force V in a beam produces shear stress; combined with flexural stress it creates **principal tensile stress** inclined at about 45 deg.
- Concrete is weak in tension, so **diagonal tension cracks** form near supports, starting at the NA and spreading to the tension face.
- Shear failure is **sudden and brittle**, hence shear steel must be provided so that the beam fails first in flexure.

Modes of shear failure

Mode	Reason	Remarks
Diagonal tension failure	a/d = 2.5 to 6, little web steel	Sudden, inclined crack to compression zone
Shear compression failure	a/d = 1 to 2.5	Crushing of concrete above the crack head
Shear tension (web crushing)	thin-webbed I beams	Crushing of web by diagonal compression

- a = shear span (distance of load from the support). Deep beams (a/d < 1) behave as **tied arch**.

Nominal shear stress

$$\tau_v = V_u / (b\ d)$$

- $b = b_w$ for flanged beams. For beams of varying depth: $\tau_v = (V_u \pm M_u \tan(\beta)/d) / (b\,d)$.
- **Critical section** for shear: at a distance **d from face of support** (when support reaction produces compression in the end region).
- The design shear strength of concrete **τ_c** depends on the percentage of tension steel $p_t = 100\,A_{st}/(b\,d)$ and f_{ck}.

Table 19 - τ_c (N/mm^2), selected values

p_t (%)	0.25	0.50	0.75	1.00	1.50	2.00	3.00 and above
M20	0.36	0.48	0.56	0.62	0.72	0.79	0.82
M25	0.36	0.49	0.57	0.64	0.74	0.82	0.92
M30	0.37	0.50	0.59	0.66	0.76	0.84	0.96

- For $p_t < 0.15$ % take τ_c for 0.15 % (M20: 0.28). **Interpolate** for intermediate p_t.
- A_{st} = area of **tension steel continuing at least d beyond the section** and properly anchored.

Table 20 - maximum shear stress $\tau_{c,max}$ (N/mm^2)

Grade	M15	M20	M25	M30	M35	M40 and above
$\tau_{c,max}$	2.5	2.8	3.1	3.5	3.7	4.0

If **$\tau_v > \tau_{c,max}$**, the section must be **redesigned** (increase b or d or grade of concrete); shear steel cannot cure web crushing.

Design of shear reinforcement (cl. 40.4)

- **Case (i): $\tau_v \le \tau_c$** - no calculation needed, but provide **minimum shear reinforcement**.
- **Case (ii): $\tau_c < \tau_v \le \tau_{c,max}$** - design stirrups for the excess shear:

$$V_{us} = V_u - \tau_c\, b\, d$$

$$s_v = 0.87 f_y A_{sv} d / V_{us}$$

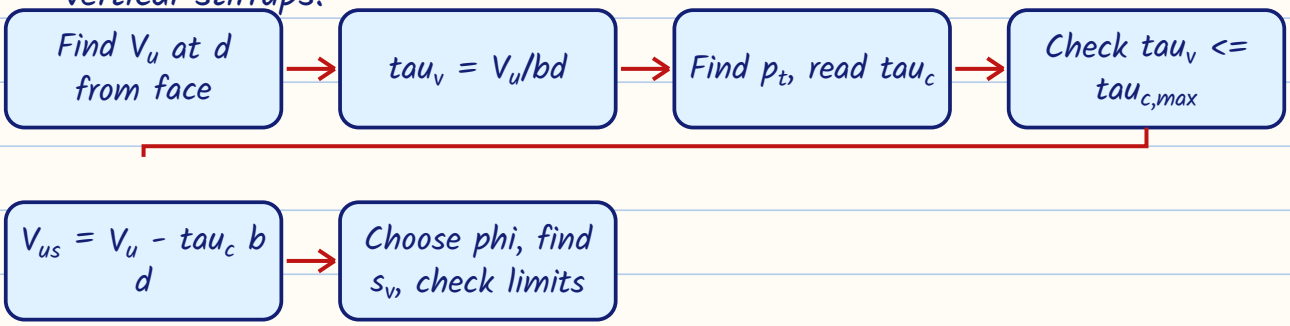
- A_{sv} = total area of stirrup legs (2-legged: 2 x pi/4 x phi^2); s_v = spacing.
- **Minimum shear reinforcement:**

$$A_{sv} / (b s_v) >= 0.4 / (0.87 f_y)$$

- **Max spacing of stirrups:** the least of **0.75 d** (vertical), **d** (inclined), **300 mm**.
- Where $\tau_v > \tau_c$, stirrup size is usually 6, 8 or 10 mm; spacing is increased towards midspan.

Bent-up bars

- Shear resistance of bent-up bars: $V_{us} = 0.87 f_y A_{sb} \sin(\alpha)$, where alpha = angle of bent (45 deg usually).
- Bent-up bars should **not carry more than 50 %** of V_{us}; the rest is by vertical stirrups.



Worked Example 3: Design of shear reinforcement

Given: simply supported beam b = 300, d = 500, M20, Fe415. Effective span 6 m, factored load w_u = 60 kN/m. Tension steel at support = 3 bars of 20 mm (942 mm^2).

- Support shear V_u = 60 x 6/2 = 180 kN. At d from face (about 0.5 m from centre): **V_u = 180 - 60 x 0.5 = 150 kN**.
- τ_v = 150 x 1000/(300 x 500) = **1.0 N/mm^2** < $\tau_{c,max}$ = 2.8, OK.
- p_t = 942 x 100/(300 x 500) = 0.63 %: τ_c = 0.48 + (0.13/0.25) x 0.08 = **0.52 N/mm^2**.

- Since $tau_v > tau_c$: $V_{us} = 150000 - 0.52 \times 300 \times 500 =$ **72 kN**.
- Use 8 mm dia 2-legged stirrups: $A_{sv} = 100.6\ mm^2$.
- $s_v = 0.87 \times 415 \times 100.6 \times 500 / 72000 =$ **252 mm**.
- Max spacing: $0.75\ d = 375$; 300; minimum steel limit $s_v <= 0.87\ f_y\ A_{sv} / (0.4\ b) = 303$ mm.

Provide 8 mm 2-legged stirrups @ 250 mm c/c near supports. In the middle region where V_u is small use the minimum shear reinforcement (@ 300 mm c/c).

5. Behaviour of RC Beams in Torsion

Basics

- **Torsion** is the twisting moment T acting about the longitudinal axis. It arises in **edge beams, spandrel beams, curved beams, canopy and balcony beams, and beams supporting eccentric loads**.
- Two types: **Equilibrium torsion** (statically determinate; must be designed for) and **compatibility torsion** (from continuity; can be redistributed, usually minimum nominal steel).
- Torsion produces **diagonal tension cracks in a spiral** around the member: it needs **closed stirrups + longitudinal bars at corners**.

Torque T → Shear flow around section → Diagonal tension spiral cracks → Closed stirrups + longitudinal bars

IS 456 method (cl. 41): equivalent shear and moment

Torsion is converted to an **equivalent shear V_e** and **equivalent bending moment M_e**, then designed as for flexure and shear.

$$V_e = V_u + 1.6\ T_u / b$$

$$M_t = T_u\ (1 + D/b) / 1.7$$

- **$M_{el} = M_u + M_t$** (for the tension face, bottom steel in sagging).
- **$M_{e2} = M_t - M_u$** (acts on the opposite face; provided only when $M_t > M_u$, steel on the compression side).
- T_u = factored torque, b = width, D = overall depth.

$$\tau_{ve} = V_e / (b\, d) \text{ must not exceed } \tau_{c,max}$$

Transverse and longitudinal reinforcement

- **Transverse (closed stirrups):**

$$A_{sv} / s_v = T_u / (b_1\, d_1\, 0.87\, f_y) + V_u / (2.5\, d_1\, 0.87\, f_y)$$

- b_1, d_1 = centre-to-centre distance between corner bars (width and depth). The total should not be less than **$(\tau_{ve} - \tau_c)\, b\, / (0.87\, f_y)$**.
- **Stirrup spacing** is the least of **x_1, $(x_1 + y_1)/4$, 300 mm** (x_1, y_1 = short and long dimensions of closed stirrup).
- **Longitudinal steel** is designed for M_{el} (bottom) and M_{e2} (top).
- **Side face steel:** for $D > 450$ mm provide **0.1 % of web area** ($b \times D$) distributed equally on two faces, spacing ≤ 300 mm.

> Torsion steel is **additional** to flexure steel but shear steel in torsion is designed using equivalent V_e; it is not added to the shear stirrups.

Worked Example 4: Beam under bending, shear and torsion

Given: beam 300 x 500 mm ($d = 450$, cover 25 mm), M25 and Fe415.

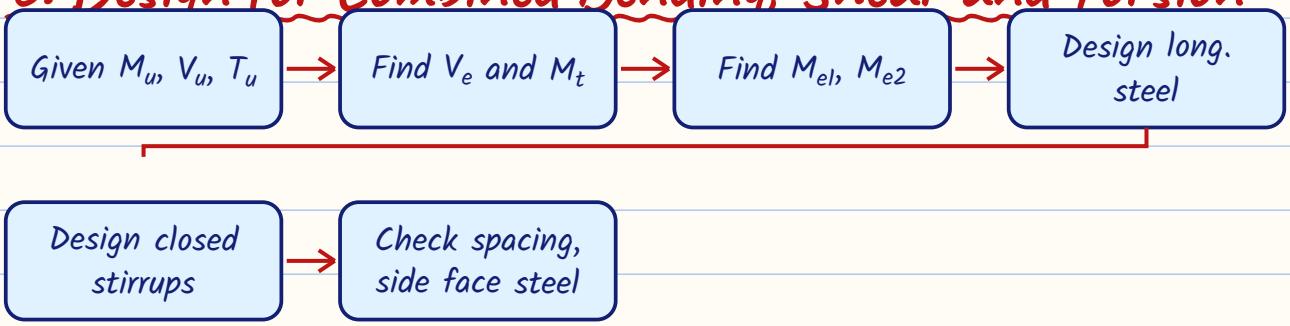
Factored loads: $M_u = 80$ kN m, $V_u = 60$ kN, $T_u = 20$ kN m.

- **V_e = 60 + 1.6 x 20/0.3 = 166.7 kN**; τ_{ve} = 166700/(300 x 450) = **1.235 N/mm²** < 3.1 (OK).
- **M_t** = 20 (1 + 500/300)/1.7 = **31.4 kN m**; M_{el} = 80 + 31.4 = **111.4 kN m**. M_{e2} = 31.4 - 80 < 0, so no top steel by torsion.
- $M_{u,lim} = 0.138 \times 25 \times 300 \times 450^2 = 210$ kN m > M_{el}: singly reinforced. $M_{el}/bd^2 = 1.83$, so **p_t = 0.56 %**, **A_{st} = 756 mm²** (provide 4-16 mm = 804 mm²).

- tau_c (p_t = 0.56 %, M25) = 0.51; minimum A_{sv}/s_v = (1.235 - 0.51) x 300/361 = **0.60 mm^2/mm**.
- b_1 = 300 - 2 x 25 - 2 x 10 - 16 = 214; d_1 = 414 mm. A_{sv}/s_v = 20 x 10^6/(214 x 414 x 361) + 60000/(2.5 x 414 x 361) = 0.625 + 0.161 = **0.79 mm^2/mm**.
- 10 mm 2-legged (157 mm^2): s_v = 199 mm; max spacing = min(214, (214 + 414)/4 = 157, 300) = 157.

Provide 10 mm 2-legged closed stirrups @ 150 mm c/c, 4-16 mm at bottom, 2-12 mm at top as hangers, and side face steel (D = 500 > 450): 0.1 % x 300 x 500 = 150 mm^2, i.e. 1-12 mm on each face.

6. Design for Combined Bending, Shear and Torsion



- A beam under **bending + shear + torsion** is designed by IS 456 cl. 41.4 using **equivalent loads**.
- **Step 1:** compute V_e and tau_{ve}; check tau_{ve} <= $tau_{c,max}$ (else redesign the section).
- **Step 2:** compute M_{el} (and M_{e2}); design the longitudinal steel.
- **Step 3:** transverse steel from torsion + shear formula; take the larger from the two checks.
- **Step 4:** detail: closed stirrups with 135 deg hooks, corner bars, side bars, correct anchorage.
- If **tau_{ve} < tau_c**, only **minimum shear reinforcement** is required, torsion steel is not needed.

7. Serviceability: Deflection and Cracking

Limit state of serviceability

- Beams must be **safe in strength** and also **serviceable**: no excessive deflection or cracking that hurts appearance or function.

Control of deflection (cl. 23.2)

- IS 456 allows the **span/effective depth ratio** check instead of calculating deflection.
- Basic values of l/d: **Cantilever 7, simply supported 20, continuous 26**. (For spans > 10 m: multiply by 10/span except cantilevers.)
- Final allowable l/d = **basic value x k_t x k_c x k_f**.

Factor	Meaning
k_t	Modification for **tension steel**; depends on percentage steel and stress $f_s = 0.58\, f_y\, (A_{st,req}/A_{st,prov})$
k_c	Modification for **compression steel** (p_c); up to 1.5
k_f	Flanged beams: 0.8 for $b_w/b_f <= 0.3$, interpolate to 1.0 at $b_w/b_f = 1$

- Total deflection limits: **span/250** (final, under all loads), and **span/350 or 20 mm** after finishing (that part affecting partitions and finishes).

Rule of thumb: if actual l/d <= allowable l/d, deflection is OK. Otherwise **increase depth d** rather than steel.

Control of cracking (cl. 26.3 and 35.3.2)

- Cracking is **inevitable** in tension zone of RC. It is controlled by limiting **crack width** and **bar spacing**.
- Limits of crack width: **0.3 mm** (normal), 0.2 mm (moderate/severe exposure) in IS 456 cl. 35.3.2.
- Use **closely spaced small diameter bars** rather than few large bars.
- Clear distance between parallel bars: not less than **bar dia, or max size of aggregate + 5 mm, or 20 mm**.
- Max horizontal spacing of tension bars: **not more than 300 mm** for

Fe250 and about **180 mm (Fe415) / 150 mm (Fe500)** per Table 15.

Quick Revision

- b_f: $T = l_o/6 + b_w + 6D_f$; $L = l_o/12 + b_w + 3D_f$.
- If $M_u <= 0.36\ f_{ck}\ b_f\ D_f\ (d - 0.42\ D_f)$, design as rectangular b_f x d.
- If $D_f/d > 0.2$, use $y_f = 0.15\ x_u + 0.65\ D_f$.
- $L_d = 0.87\ f_y\ phi\ 1/(4\ tau_{bd})$; check $L_d <= 1.3\ M_1/V + L_o$.
- Shear: $tau_v = V_u/bd$; $V_{us} = V_u - tau_c\ b\ d$; $s_v = 0.87\ f_y\ A_{sv}\ d/V_{us}$.
- Torsion: $V_e = V_u + 1.6\ T/b$; $M_t = T(1 + D/b)/1.7$; $M_{el} = M_u + M_t$.
- Deflection: l/d = 7, 20, 26 x modification factors.

Important Questions

- 1. Derive the expression for M_u of a T-beam when the NA is in the web and $D_f/d > 0.2$.
- 2. Explain effective flange width of T and L beams as per IS 456.
- 3. Define bond. Explain development length and the check at simple supports.
- 4. Explain the modes of shear failure of RC beams. How is shear reinforcement designed?
- 5. A T-beam has $b_f = 1200$, $b_w = 250$, $D_f = 100$, $d = 500$. Find A_{st} for $M_u = 350$ kN m.
- 6. Explain the IS 456 method of design for combined bending, shear and torsion.
- 7. Write notes on the control of deflection and cracking in beams.

UNIT - III
Limit State Design of Slabs

Design of Reinforced Concrete Structures (23A01502) - handwritten study notes

1. Introduction and Classification of Slabs

- A **slab** is a plane structural element, thin in comparison with its other two dimensions. It carries transverse loads (floor, roof, live load) mainly by **bending** and transfers them to beams, walls or columns.
- Design is done by the **Limit State Method** of **IS 456:2000**. Concrete M20 and steel Fe415 are used in the examples below.
- Slabs are classified by **support condition** and by **direction of load transfer**.

One-way slab

- Supported on two opposite sides only, **or** supported on four sides with ratio $l_y/l_x > 2$.
- Main steel is provided in the **short span** direction. Distribution (temperature) steel is provided in the long direction.
- Slab bends in **one direction** - designed as a 1 m wide beam strip.

Two-way slab

- Supported on all four sides with $l_y/l_x <= 2$. Bending takes place in **both directions**.
- Main steel is provided in both directions; the short span carries a larger share of the load.

Rule: $l_y / l_x > 2 \longrightarrow$ One-way slab. $l_y / l_x <= 2 \longrightarrow$ Two-way slab. Here l_x = short span, l_y = long span.

Feature	One-way slab	Two-way slab
Supports	Two opposite sides (or ratio > 2)	Four sides, ratio <= 2

Bending	In short direction only	In both directions
Main steel	One direction	Both directions
Distribution steel	Required in long direction	Not required (main steel both ways)
Torsion steel at corners	Not needed	Needed at discontinuous corners

Loads on slabs

- **Dead load**: self weight = D x 25 kN/m^3, floor finish (about 1.0 kN/m^2), partitions.
- **Imposed (live) load** as per IS 875 Part 2: residences 2 kN/m^2, offices 2.5 - 4 kN/m^2, balconies 3 kN/m^2, classrooms 3 kN/m^2.
- **Factored load** $w_u = 1.5\,(DL + LL)$.

2. Basic Design Data (IS 456:2000)

Effective span (Cl. 22.2)

- **Simply supported slab**: the smaller of (i) clear span + effective depth d, (ii) centre to centre of supports.
- **Cantilever**: length to the face of support + d/2 (cantilever which is the end of a continuous slab: to the centre of support).
- **Continuous slab**: centre to centre of supports (if support width is small compared to span).

Minimum depth: span / effective depth ratio (Cl. 23.2.1)

Support condition	Basic l/d
Cantilever	7
Simply supported	20
Continuous	26

- Basic value is **multiplied by a modification factor** k_t from Fig. 4 of IS 456, which depends on the percentage steel p_t and the service stress f_s.

- Service stress $f_s = 0.58\, f_y \times (A_{st\ required} / A_{st\ provided})$.
- For two-way slabs up to 3.5 m span with live load up to 3 kN/m²: **simply supported 35**, **continuous 40** for mild steel; multiply by **0.8** for Fe415 (28 and 32).
- For spans over 10 m the values are multiplied by 10 / span (not for cantilevers).

Flexure formulae (singly reinforced, per 1 m width)

$$M_{u,lim} = 0.138\, f_{ck}\, b\, d^2 \quad (Fe415)$$

$$M_u = 0.87\, f_y\, A_{st}\, d\, [1 - A_{st}\, f_y / (b\, d\, f_{ck})]$$

$$A_{st} = (0.5\, f_{ck} / f_y)\, [1 - \mathrm{sqrt}(1 - 4.6\, M_u / (f_{ck}\, b\, d^2))]\, b\, d$$

- Slabs are always designed **under-reinforced** ($x_u < x_{u,max}$); doubly reinforced slabs are not normally needed.
- Overall depth D = d + clear cover + half the bar diameter. Minimum **clear cover 20 mm** (mild exposure); 15 mm may be used for D < 100 mm in some cases.

Reinforcement limits

- **Minimum steel** (Cl. 26.5.2.1): 0.15 % of b D for mild steel (Fe250); **0.12 %** of b D for HYSD bars (Fe415, Fe500).
- **Maximum bar diameter** <= D / 8.
- **Spacing of main steel**: not more than **3d** or **300 mm**, whichever is smaller.
- **Spacing of distribution steel**: not more than **5d** or **450 mm**, whichever is smaller.
- Minimum bar diameter used is generally 8 mm for main steel, 6 or 8 mm for distribution steel.

> Remember: main bars ⟶ 3d / 300 mm. Distribution bars ⟶ 5d / 450 mm. Minimum steel ⟶ 0.12 % (HYSD) or 0.15 % (mild steel) of bD.

Shear in slabs (Cl. 40.2)

- Nominal shear stress: $\tau_v = V_u / (b\ d)$.
- Slabs are generally safe in shear; no shear reinforcement is provided, the slab is safe if $\tau_v < k\ \tau_c$.
- τ_c is taken from **Table 19** (depends on p_t and grade of concrete). For M20: p_t 0.15 ⟶ 0.28, 0.25 ⟶ 0.36, 0.50 ⟶ 0.48, 0.75 ⟶ 0.56, 1.0 ⟶ 0.62 N/mm^2.
- The factor **k** depends on overall depth (Cl. 40.2.1.1):

Overall depth D (mm)	300 or more	275	250	225	200	175	150 or less
k	1.00	1.05	1.10	1.15	1.20	1.25	1.30

- Maximum shear stress $\tau_{c,max}$ for M20 = 2.8 N/mm^2; if exceeded, increase depth.

3. Design of Cantilever Slab

- Cantilever slabs have **hogging moment**; main steel is placed at the **top** (tension face) and extends well into the backing slab/beam.
- **Design moment**: $M_u = w_u L^2 / 2$, shear $V_u = w_u L$ (at the support).
- The slab is usually tapered (thicker at the support, thinner at the free end): depth at the support = L / 7 x k_t approximately.
- Main steel should be **anchored** into the support by at least L_d and preferably continued into the adjacent span.
- Distribution steel is placed under the main steel (at right angles) at 0.12 % of bD.

Assume depth d = L / (7 k_t) → Load w_u (incl. parapet) → $M_u = w_u L^2/2$ → A_{st} at top

Distribution steel → Check shear and deflection

Example 1: Cantilever slab (balcony)

Problem: Design a cantilever balcony slab of projection 1.5 m. Live load 3 kN/m^2, floor finish 1 kN/m^2. Use M20 and Fe415.

- **Depth**: take D = 180 mm, d = 180 - 20 - 5 = 155 mm. Check: L/d = 1500 / 155 = 9.7 < 7 x k_t (with k_t around 1.8, allowable 12.6).
- **Loads**: self weight 0.18 x 25 = 4.5, finish 1.0, live 3.0. Total = 8.5 kN/m^2. **w_u = 1.5 x 8.5 = 12.75 kN/m**.
- **M_u** = 12.75 x 1.5^2 / 2 = **14.34 kNm**; V_u = 12.75 x 1.5 = 19.1 kN.
- Check depth: d = sqrt(14.34 x 10^6 / (0.138 x 20 x 1000)) = 72 mm < 155 mm. So the section is under-reinforced.
- **Steel**: M_u / (b d^2) = 14.34 x 10^6 / (1000 x 155^2) = 0.597. A_{st} = 0.024 x [1 - sqrt(1 - 4.6 x 0.597 / 20)] x 1000 x 155 = **266 mm^2**.
- Use 10 mm bars: spacing = 78.5 x 1000 / 266 = 295 mm. Provide **10 mm @ 275 mm c/c** (A_{st} = 285 mm^2) at top. Spacing < 3d = 465 mm. OK.
- **Distribution steel** = 0.12 % x 1000 x 180 = 216 mm^2. Provide **8 mm @ 230 mm c/c** (219 mm^2).
- **Shear**: tau_v = 19100 / (1000 x 155) = 0.123 N/mm^2. p_t = 100 x 285 / (1000 x 155) = 0.184 %, tau_c = 0.30. k = 1.24 ⟶ k tau_c = 0.37 > 0.123. **Safe**.
- **Deflection**: f_s = 0.58 x 415 x (266/285) = 225 N/mm^2; k_t about 1.8. Allowable l/d = 7 x 1.8 = 12.6 > 9.7. **Safe**.

Ans: D = 180 mm; main steel 10 mm @ 275 c/c on top, extended into the support by L_d; distribution 8 mm @ 230 c/c.

4. One-Way Simply Supported Slab

- Design a **1 m wide strip** as a simply supported beam of span l_{eff}.

$$M_u = w_u l^2 / 8 \quad V_u = w_u l / 2$$

- The reinforcement bars are placed at the **bottom** and **spaced** such that spacing <= 3d and 300 mm.
- **Alternate bars** may be bent up near the support at about l/7 from the face (for continuity and anchorage); at least **one third** of the positive steel must be extended into the support (Cl. 26.2.3.3).
- At a simple support the bars must be anchored: $M_1 / V + L_0 >= L_d$, where L_0 = embedment beyond centre of support (not more than d or 12 diameters).
- **Development length** $L_d = 0.87 f_y \text{ phi} / (4 \tau_{bd})$. For M20 and Fe415 ($\tau_{bd}$ = 1.2 x 1.6 = 1.92 N/mm²): **L_d = 47 phi**.

Example 2: One-way simply supported slab

Problem: Design a one-way slab for a room with clear span 3.5 m, walls 230 mm thick on both sides. Live load 4 kN/m², floor finish 1 kN/m². M20, Fe415.

- **Depth**: $l/d = 20 \times k_t$; assume $k_t = 1.4 \longrightarrow d = 3600 / 28 = 129$ mm. Provide D = 170 mm, d = 170 - 25 - 5 = **140 mm** (12 mm bars, 20 mm cover).
- **Effective span**: clear + d = 3.5 + 0.14 = 3.64 m; c/c = 3.5 + 0.23 = 3.73 m. **l_{eff} = 3.64 m**.
- **Loads**: self weight 0.17 x 25 = 4.25, finish 1.0, live 4.0 $\longrightarrow$ 9.25 kN/m². **w_u = 1.5 x 9.25 = 13.875 kN/m**.
- **M_u** = 13.875 x 3.64² / 8 = **22.98 kNm**; **V_u** = 13.875 x 3.64 / 2 = **25.25 kN**.
- Depth check: $d_{min} = \text{sqrt}(22.98 \times 10^6 / 2760) = 91$ mm < 140 mm. OK.
- **Steel**: $M_u / (b d^2) = 22.98 \times 10^6 / (1000 \times 140^2) = 1.172$. A_{st} = 0.024096 x [1 - sqrt(1 - 4.6 x 1.172 / 20)] x 1000 x 140 = **490 mm²**.

- 12 mm bars: spacing = 113 x 1000 / 490 = 230 mm. Provide **12 mm @ 230 mm c/c** (A_{st} = 491 mm^2). 230 < 3d = 420 and < 300. OK.
- **Distribution steel** = 0.12 % x 1000 x 170 = 204 mm^2: **8 mm @ 240 mm c/c** (209 mm^2) < 5d = 700 and 450. OK.
- **Shear**: tau_v = 25250 / (1000 x 140) = 0.18 N/mm^2; p_t = 0.35 %, tau_c = 0.40; k = 1.25 ⟶ k tau_c = 0.50 > 0.18. **Safe**.
- **Deflection**: f_s = 0.58 x 415 x 490/491 = 240; p_t = 0.35 ⟶ k_t about 1.4 (Fig. 4). Allowable l/d = 20 x 1.4 = 28; actual = 3640 / 140 = 26. **Safe**.
- **Development length**: L_d = 47 x 12 = 564 mm. At support: M_1 / V = 0.87 x 415 x 491 x 140 / 25250 = 983 mm, so anchorage is OK.

Ans: D = 170 mm; main steel 12 mm @ 230 c/c; distribution 8 mm @ 240 c/c; bars bent up alternately near the support.

5. Continuous Slabs

- Continuous slabs are economical: smaller moments and smaller depth than simply supported slabs. Spans may be taken as c/c of supports.
- **IS 456 Table 12** gives bending moment coefficients and **Table 13** gives shear force coefficients for continuous beams and slabs of **three or more equal spans** (or spans differing by not more than 15 %), carrying uniformly distributed loads, where the imposed load does not exceed 3/4 of the dead load (Cl. 22.5.1).

Table 12: Bending moment = coefficient x w l^{2}

Type of load	Mid of end span	Support next to end support	Mid of interior span	Other interior supports
Dead load (fixed)	+1/12	-1/10	+1/16	-1/12
Imposed load (not fixed)	+1/10	-1/9	+1/12	-1/9

Table 13: Shear force = coefficient x w l

Type of load	End support	Next to end - outer side	Next to end - inner side	Other supports
Dead load (fixed)	0.4	0.6	0.55	0.5
Imposed load (not fixed)	0.45	0.6	0.6	0.6

- Moment for each load case = (coefficient for DL) x $w_d l^2$ + (coefficient for LL) x $w_l l^2$, where w_d and w_l are the **factored** dead and imposed loads.
- The effective span l is the **centre to centre** distance; for two spans only, or unequal spans, moment distribution should be done.
- Design the section for the **maximum moment** (usually the support next to the end support); use the same depth throughout.
- **Detailing**: positive steel at the bottom in the span, negative steel on top over the supports; **bent-up bars** in spans are carried over the support (alternate bars bent up and extended as top steel).

Example 3 (outline): Continuous slab with three equal spans of 3.5 m

- Take D = 130 mm, d = 105 mm. Dead load = 0.13 x 25 + 1.0 = 4.25 kN/m^2; live load = 4 kN/m^2.
- Factored w_d = 1.5 x 4.25 = 6.375 kN/m; w_l = 1.5 x 4 = 6.0 kN/m (per metre width). l^2 = 12.25.

Section	Dead load part	Imposed load part	Total M_u (kNm)
Mid of end span	6.375 x 12.25 / 12 = 6.51	6.0 x 12.25 / 10 = 7.35	+13.86
Support next to end support	6.375 x 12.25 / 10 = 7.81	6.0 x 12.25 / 9 = 8.17	-15.98
Mid of interior span	6.375 x 12.25 / 16 = 4.88	6.0 x 12.25 / 12 = 6.13	+11.01

Other interior supports	6.375 x 12.25 / 12 = 6.51	6.0 x 12.25 / 9 = 8.17	-14.68

- **Maximum shear** (support next to end, outer side): 0.6 x (6.375 + 6.0) x 3.5 = **26.0 kN**.
- Depth check: d_{min} = sqrt(15.98 x 10^6 / 2760) = 76 mm < 105 mm. OK.
- A_{st} at the support next to end support = **464 mm²** (M_u / b d^2 = 1.449) ⟶ 10 mm @ 160 mm c/c (491 mm^2) top steel. Other sections are designed in the same way. Distribution steel 0.12 % x 1000 x 130 = 156 mm^2 ⟶ 8 mm @ 300 c/c.

Remember: for continuous slabs read coefficients separately for dead and imposed load; hogging moments are larger than sagging moments.

6. Two-Way Slabs - IS 456 Annex D

- Two-way slabs are designed by the **coefficient method** of **Annex D** (Cl. 24.4).
- **Annex D-1**: simply supported slabs with corners **free to lift** - Table 27.
- **Annex D-2**: slabs **restrained** at the edges (corners held down) - Table 26.

$$M_x = \text{alpha}_x\, w_u\, l_x^2 \quad M_y = \text{alpha}_y\, w_u\, l_x^2$$

- l_x = shorter effective span; l_y = longer effective span; M_x acts on the strip of span l_x, M_y on the strip of span l_y. Both moments are per metre width.
- alpha_x, alpha_y are moment coefficients that depend on l_y / l_x and the **edge condition**.

Nine edge conditions in Table 26

- **Case 1** interior panel; **2** one short edge discontinuous; **3** one long edge discontinuous; **4** two adjacent edges discontinuous.

- **Case 5** two short edges discontinuous; **6** two long edges discontinuous; **7** three edges discontinuous (one long edge continuous); **8** three edges discontinuous (one short edge continuous); **9** four edges discontinuous.

Selected coefficients (Table 26) for alpha_{x}

Case	Moment	l_y/l_x = 1.0	1.5	2.0
1 Interior	Negative	0.032	0.053	0.065
1 Interior	Positive	0.024	0.041	0.049
4 Two adjacent edges disc.	Negative	0.047	0.075	0.091
4 Two adjacent edges disc.	Positive	0.035	0.056	0.069
9 Four edges disc.	Positive	0.056	0.089	0.107

- For case 1: **$alpha_y$** = 0.032 (negative) and 0.024 (positive) for all values of l_y/l_x. For case 4 $alpha_y$ = 0.047 (negative) and 0.035 (positive).
- Read other values from **Table 26** of IS 456 and interpolate for intermediate ratios.

Table 27: Simply supported slab, corners not held down

l_y/l_x	1.0	1.1	1.2	1.3	1.4	1.5	1.75	2.0
$alpha_x$	0.062	0.074	0.084	0.093	0.099	0.104	0.113	0.118
$alpha_y$	0.062	0.061	0.059	0.055	0.051	0.046	0.037	0.029

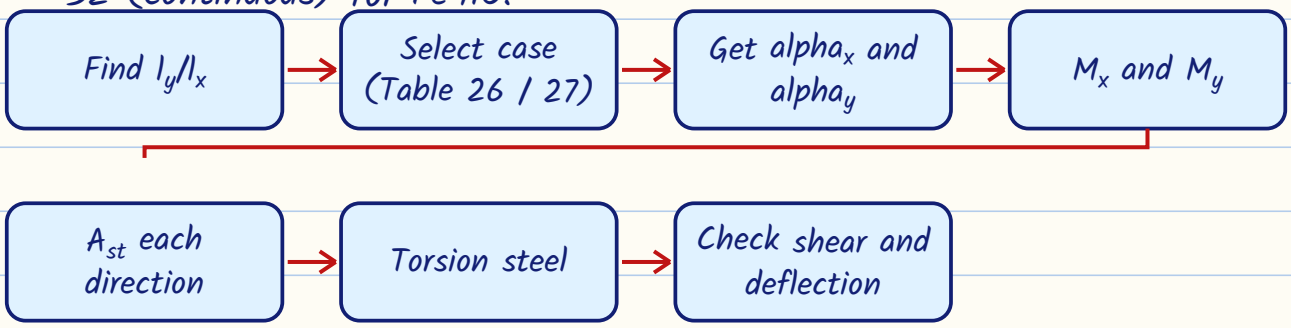
Torsion reinforcement at corners (Cl. D-1.8)

- When corners are **prevented from lifting**, torsion steel is provided at the corners where the slab is discontinuous.
- **Corner with both edges discontinuous**: area of steel in each of the four layers (top and bottom, in both directions) = **3/4 of the area required for the maximum mid-span moment** (short span).
- **Corner with one edge continuous and the other discontinuous**: **half** of the above (3/8 of the mid-span area).
- **No torsion steel** where the slab is continuous over both edges.
- The torsion steel extends from the edge a distance of **l_x / 5** in each

direction.

Other design rules for two-way slabs

- The slab is divided into a **middle strip** (3/4 of the width) and **two edge strips** (each 1/8 of the width) in each direction. The middle strip carries the full moments; edge strips need only **minimum steel** (0.12 %).
- Short-span bars are placed **nearer to the tension face** (outer layer, larger d); long-span bars lie above them with d reduced by one bar diameter: $d_y = d_x - phi$.
- **Discontinuous edges**: negative moment at the discontinuous edge = 1/3 of the mid-span moment (or as given in the table).
- Bar spacing limits: 3d / 300 mm for main steel in both directions.
- **Shear**: $V_u = w_u l_x / 2$ per metre width (check $tau_v < k\ tau_c$); the load is shared between the two directions, so the short-span value is conservative.
- **Deflection**: use l_x / d with the basic values 28 (simply supported) and 32 (continuous) for Fe415.



Remember: two-way slab ⟶ short span steel is the bottom layer; torsion steel = 3/4 of mid span steel (two discontinuous edges) and 1/2 of that for one discontinuous edge; length $l_x/5$.

Example 4: Restrained two-way slab (corners held down)

Problem: A slab panel 4 m x 5 m (clear spans) is discontinuous along two adjacent edges and continuous on the other two edges. Live load 3 kN/m^2, floor finish 1 kN/m^2, M20, Fe415. Design the slab.

- $l_y / l_x = 5 / 4 = 1.25 \longrightarrow$ Case 4 (two adjacent edges discontinuous). Coefficients (interpolated for 1.25): $alpha_x(-) = 0.0625$, $alpha_x(+) = 0.047$, $alpha_y(-) = 0.047$, $alpha_y(+) = 0.035$.
- **Depth**: l/d = 28 (Fe415, simply supported) x k_t > 4000 / 28 = 143 mm; for continuous slabs a smaller depth can be accepted. Take **D = 150 mm**, d_x = 125 mm, d_y = 115 mm (10 mm bars).
- **Load**: self weight 0.15 x 25 = 3.75 + 1.0 + 3.0 = 7.75 kN/m^2; **w_u = 11.625 kN/m^2**. $w_u l_x^2$ = 11.625 x 16 = 186 kNm.

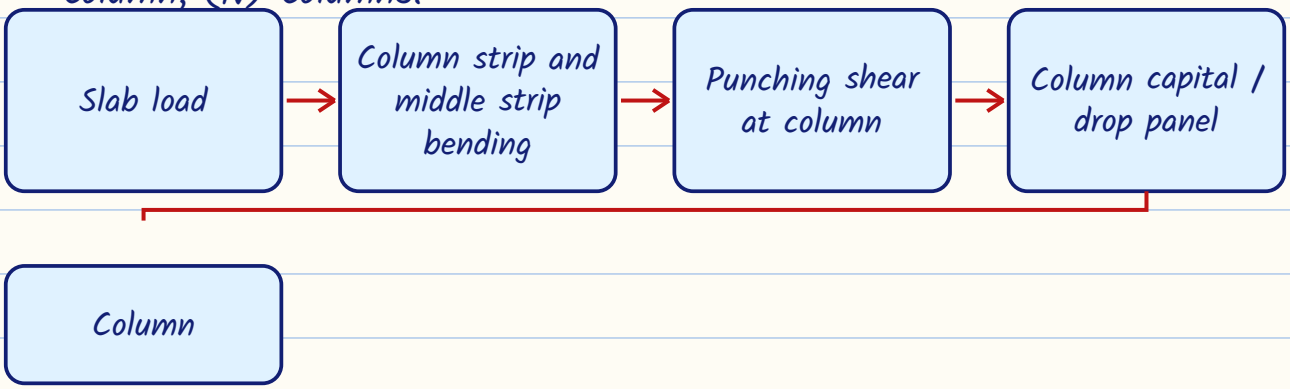
Moment	Coefficient	M_u (kNm/m)	d (mm)	A_{st} req. (mm^2)	Provide
M_x negative	0.0625	11.63	125	270	10 mm @ 280
M_x positive	0.047	8.74	125	200	8 mm @ 240
M_y negative	0.047	8.74	115	219	8 mm @ 220
M_y positive	0.035	6.51	115	162 (min 180)	8 mm @ 270

- Minimum steel = 0.12 % x 1000 x 150 = **180 mm^2**. Spacing < 3d = 375 mm and < 300 mm in all cases.
- **Torsion steel** at the corner where two adjacent edges are discontinuous: 3/4 x 200 = **150 mm^2** in each of four layers $\longrightarrow$ 8 mm @ 300 mm c/c, extended over l_x/5 = 4000 / 5 = **800 mm** from the edges. At corners with one discontinuous edge: half of this.
- **Shear**: V_u = 11.625 x 4 / 2 = 23.25 kN/m; tau_v = 23250 / (1000 x 125) = 0.186 N/mm^2. p_t = 100 x 270 / (1000 x 125) = 0.22 %, tau_c about 0.33, k = 1.3 $\longrightarrow$ 0.43 > 0.186. **Safe**.
- **Deflection**: l/d = 4000 / 125 = 32; allowable = 28 x k_t (k_t about 1.3 $\longrightarrow$ 36). **Safe**.

8. Introduction to Flat Slabs

- A **flat slab** is a reinforced concrete slab supported **directly on columns** without beams. The loads are transferred to the columns by bending and shear (punching) in the slab.

- **Components**: (i) slab, (ii) **drop panel** - thickened part of the slab around the column, (iii) **column capital** (head) - flared top of the column, (iv) columns.



Drop panel and column capital (IS 456 Cl. 31)

- **Drop panel**: should extend in each direction at least **1/3 of the panel length** in that direction; its projection below the slab should not be more than **D/4**; the total depth used in the design near the column is limited to slab thickness + D/4.
- **Column capital**: is designed for the effective diameter measured where the 45 degree line from the capital meets the soffit of the slab (or drop). The usual dimension is about 0.25 l for the capital diameter.
- **Minimum thickness** of a flat slab = **125 mm**. Span/depth ratio: l/d as per Cl. 23.2 (for flat slabs the longer span is used; multiply by 0.9 when there is no drop panel).

Column strip and middle strip

- **Column strip**: width on each side of the column centreline = 0.25 x the **smaller** of l_1 and l_2.
- **Middle strip**: the remaining part of the panel between two column strips.
- Moments are first found for the whole panel and then divided between column strip and middle strip.

Direct design method (Cl. 31.4)

- Conditions: at least **three continuous spans** in each direction; ratio of longer to shorter span not more than 2; successive span lengths differ

by not more than 1/3 of the longer span; column offset not more than 10 % of span; **live load not more than 3 times the dead load**; loads uniformly distributed.

$$M_o = W l_n / 8 \quad W = w_u l_2 l_n$$

- l_n = clear span in the direction of moments, l_2 = span transverse to l_1.
- **Interior panel**: negative design moment = **0.65 M_o**, positive design moment = **0.35 M_o**.

Moment	Column strip (% of panel moment)	Middle strip
Negative (at support)	75 %	25 %
Positive (mid span)	60 %	40 %

Punching (two-way) shear check (Cl. 31.6)

- The critical section is at a distance **d/2 from the periphery of the column / capital / drop**.
- Perimeter $b_o = 2 (c_1 + d) + 2 (c_2 + d)$ for a rectangular column of sides $c_1 \times c_2$.

$$tau_v = V_u / (b_o d) <= k_s tau_c$$

- tau_c = **0.25 sqrt(f_{ck})** (= 1.12 N/mm^2 for M20) and **k_s = 0.5 + $beta_c$** (not more than 1), where $beta_c$ = ratio of short side to long side of the column.
- If the check fails: increase thickness, provide a **drop panel**, a larger column capital, or shear reinforcement (stirrups / shear heads).

Advantages and limitations of flat slabs

Advantages	Limitations
Flat soffit - simple formwork	Higher punching shear risk at columns
Less storey height / better headroom	Thicker slab (more concrete) for the same span

Remember: drop panel >= l/3, projection <= D/4, punching at d/2 from column, tau_c = 0.25 sqrt(f_{ck}), min thickness 125 mm.

Quick Revision

- One-way: l_y/l_x > 2; two-way: <= 2. Main steel in short span; distribution 0.12 %.
- Basic l/d: **7 (cantilever), 20 (simply supported), 26 (continuous)** x k_t.
- Shear check: tau_v < k tau_c; k = 1.3 for D <= 150 down to 1.0 for D >= 300.
- Continuous slab: Table 12 (moments) and Table 13 (shears); live load <= 3/4 dead load.
- Two-way: M_x = $alpha_x$ w l_x^2, M_y = $alpha_y$ w l_x^2; Table 26 (restrained) and Table 27 (simply supported).
- Torsion steel: 3/4 of mid span (two discontinuous edges), 1/2 (one), none (continuous); length l_x/5.
- Flat slab: drop panel l/3, capital, column strip 0.25 l, M_o = W l_n/8, 0.65 / 0.35 split, punching at d/2.

Important Questions

- 1. Design a cantilever slab of 1.2 m projection for a live load of 3 kN/m^2. Use M20 and Fe415.
- 2. Design a one-way simply supported slab for a room 3 m x 8 m (clear), live load 3 kN/m^2, with detailing sketch.
- 3. State the coefficients of IS 456 Tables 12 and 13 for continuous slabs. Design an end span of a continuous slab.
- 4. Design a two-way slab 4 m x 5 m restrained at all edges with two adjacent edges discontinuous. Show the torsion reinforcement.
- 5. Write short notes on flat slabs: drop panel, column capital, column strip, punching shear and direct design method.

UNIT - IV
Limit State Design of Columns & Footings

Design of Reinforced Concrete Structures (23A01502) - handwritten study notes

Syllabus Topics

- **Columns**: types of columns, design of short rectangular and circular columns for axial load, uniaxial bending and biaxial bending.
- **Footings**: wall footing, axially and eccentrically loaded rectangular pad and sloped footings, combined rectangular footing for two columns only.
- Code used: **IS 456 : 2000** (columns - clause 25, 26, 39; footings - clause 34) and **SP-16** design charts.

Exam tip: Unit IV always carries one column question (axial / uniaxial with SP-16 chart) and one footing question (isolated or combined). Learn the steps + one worked example of each.

1. Columns - Introduction

Definition

- A **column** is a vertical compression member whose effective length is more than **3 times** its least lateral dimension (otherwise it is a pedestal).
- It transfers the load of beams / slabs above to the foundation. Failure of a column is **sudden and brittle**, so a higher factor of safety is used.
- Material safety factors: concrete 1.5, steel 1.15; load factor 1.5 (DL + LL).

Classification of Columns

(a) Based on reinforcement and shape

- **Tied column**: longitudinal bars + lateral ties (rectangular, square, L-shaped). Most common.

- **Spiral (helically reinforced) column**: longitudinal bars + continuous helix, usually circular. More ductile and tough, takes 5% extra load.
- **Composite column**: structural steel section (or pipe) encased in or filled with concrete.

(b) Based on slenderness ratio

- **Short column**: $l_{ex}/D < 12$ and $l_{ey}/b < 12$ - fails by crushing of concrete / yielding of steel (material failure).
- **Long (slender) column**: ratio >= 12 - fails by **buckling**; additional moments M_a are to be added (clause 39.7).

(c) Based on loading

- **Axially loaded column** (theoretical, eccentricity only minimum e_{min}).
- **Uniaxial bending**: axial load + moment about one axis.
- **Biaxial bending**: axial load + moments about both axes (corner columns).

Effective Length (Table 28, IS 456)

- Effective length $l_e = k \times l$ (unsupported length). Depends on end conditions:

End condition	l_e / l
Fixed at both ends	0.65
Fixed one end, hinged other	0.80
Hinged both ends	1.00
Fixed one end, free other (cantilever)	2.00

- **Slenderness ratio** = l_e / D (D = depth in the direction considered). If $l_e/D < 12$, the column is short.

Minimum Eccentricity (clause 25.4)

$$e_{min} = l/500 + D/30 >= 20 \text{ mm}$$

- l = unsupported length of the column, D = lateral dimension in the direction under consideration.

- Every column must be designed for at least this eccentricity, in each direction.
- If e_{min} <= 0.05 D, the column can be designed for **axial load only** with the formula below.

For biaxial design, e_{min} is checked **separately** about each axis; but it is not applied simultaneously.

2. Code Requirements for Columns

Longitudinal Steel (clause 26.5.3.1)

- Minimum steel = **0.8%** of gross area A_g; maximum = **6%** (practically **4%** to ease concreting, lap zones need 6% max).
- Minimum **4 bars** in rectangular columns and **6 bars** in circular columns.
- Minimum bar diameter = **12 mm**.
- Cover to column bars: **40 mm** or the bar dia, whichever more (nominal cover 40 mm, 25 mm for small columns <= 200 mm and bars <= 12 mm).

Transverse Reinforcement (clause 26.5.3.2)

- **Lateral ties**: dia >= **1/4 of largest longitudinal bar** and not less than **6 mm**.
- **Pitch of ties** = least of:
  - least lateral dimension of column
  - 16 x dia of smallest longitudinal bar
  - 300 mm

Helical Reinforcement (clause 26.5.3.2 d)

- Pitch <= **75 mm** and <= core dia / 6; pitch >= **25 mm** and >= 3 x dia of helix.

3. Axially Loaded Short Columns (clause 39.3)

$$P_u = 0.4\, f_{ck}\, A_c + 0.67\, f_y\, A_{sc}$$

- P_u = factored axial load, A_c = net concrete area = $A_g - A_{sc}$, A_{sc} = area of longitudinal steel.
- The 0.4 and 0.67 factors already include the **minimum eccentricity** effect (0.45 f_{ck} and 0.75 f_y are the pure axial strengths, reduced by about 10% for eccentricity).
- **Helically reinforced** short columns: take **1.05 times** the strength:

$$P_u = 1.05\,(\,0.4\, f_{ck}\, A_c + 0.67\, f_y\, A_{sc}\,)$$

Design Steps - Axially Loaded Column

Factored load P_u → Assume p (1 to 2%) → Find A_g → Fix b and D

Find A_{sc} (→ = 0.8%) → Ties / detailing

Worked Example 1 - Square tied column

Q. Design a short square column to carry a **service load of 1000 kN**. Use M25 and Fe 415. Unsupported length 3.5 m.

- $P_u = 1.5 \times 1000 =$ **1500 kN**.
- Assume p = 1.5%: $1500 \times 10^3 = A_g\,[\,0.4 \times 25 \times 0.985 + 0.67 \times 415 \times 0.015\,] = A_g\,(9.85 + 4.17) = 14.02\, A_g$
- $A_g = 107{,}000\ mm^2 \longrightarrow$ provide **350 x 350 mm** ($A_g = 122{,}500\ mm^2$).
- Slenderness: $l_e/D = 3500 / 350 = 10 < 12$ so short. $e_{min} = 3500/500 + 350/30 = 18.7$ mm, i.e. less than 20 mm, so $e_{min} = 20$ mm. Since $e_{min} > 0.05\ D = 17.5$ mm, the 0.4 / 0.67 formula (which already allows for e_{min}) is used.
- $1500 \times 10^3 = 0.4 \times 25\,(122500 - A_{sc}) + 0.67 \times 415\, A_{sc} = 1{,}225{,}000 +$

268.05 A_{sc}

- A_{sc} = 275,000 / 268.05 = **1026 mm²** (min 0.8% = 980 mm², OK)
- Provide **4 - 20 mm dia** (1257 mm², p = 1.03%).
- Check: P_u = 0.4 x 25 x 121,243 + 0.67 x 415 x 1257 = 1212 + 350 = **1562 kN > 1500 kN, safe**.
- Ties: dia >= 20/4 = 5 mm and >= 6 mm ⟶ **8 mm** (use 8 mm as per practice). Pitch = least of (350, 16 x 20 = 320, 300) = **300 mm**.

Final: 350 x 350 column, 4 - 20 mm bars, 8 mm ties @ 300 mm c/c.

Worked Example 2 - Circular column with helical ties

Q. A circular column with helical reinforcement carries a factored load of **1800 kN**. M25, Fe 415. Design the column.

- Strength per unit: P_u/1.05 = 1800 / 1.05 = 1714 kN.
- Assume p = 1.5%: 1714 x 10^3 = 14.02 A_g ⟶ A_g = 122,270 mm².
- D = sqrt(4 A_g / pi) = sqrt(4 x 122270 / 3.1416) = 394.6 mm ⟶ provide **D = 400 mm**; A_g = 125,664 mm².
- 1,714,286 = 0.4 x 25 (125664 - A_{sc}) + 278.05 A_{sc} = 1,256,640 + 268.05 A_{sc}
- A_{sc} = 457,646 / 268.05 = **1707 mm²**, min 0.8% = 1005 mm² OK.
- Provide **6 - 20 mm** bars (1885 mm²).
- Helix: dia = 20/4 = 5 ⟶ use **8 mm**. Core dia = 400 - 2 x 40 = 320 mm; pitch <= core dia/6 = 53 mm; <= 75 mm; >= 25 mm and >= 3 x 8 = 24 mm. Provide **8 mm helix @ 50 mm pitch**.

Final: D = 400 mm, 6 - 20 mm bars, 8 mm helix at 50 mm pitch.

4. Columns under Uniaxial Bending (SP-16 charts)

- Columns in frames carry **P_u + M_u** about one axis. The exact analysis uses strain compatibility, stress block and the interaction diagram. SP-16 gives ready charts (Charts 27 to 62).

Interaction Diagram (P_u - M_u)

- Plot of axial capacity against moment capacity for a given section and steel.
- Points: pure axial (P_{uz} at zero moment), **balanced failure** (compression and tension failure together), pure bending (zero axial load).

Chart Parameters (non-dimensional)

$$P_u / (f_{ck}\, b\, D) \text{ and } M_u / (f_{ck}\, b\, D^2)$$

- d' = effective cover to the centre of the compression bars; ratio **d'/D** = 0.05, 0.10, 0.15, 0.20 gives separate charts.
- The chart gives **p/f_{ck}**, where $p = 100\, A_{sc} / (b\, D)$. So $p = (p/f_{ck}) \times f_{ck}$.

Design Procedure

- For long columns add the additional moment $M_a = P_u\, D\, (l_e/D)^2 / 2000$.

Worked Example 3 - Uniaxial bending

Q. A column 300 x 500 mm (D = 500 mm) is subjected to P_u = 1200 kN and M_u = 150 kN.m about the major axis. Effective length 3.5 m. M25, Fe 415, d' = 50 mm. Find A_{sc}.

- $e_{min} = 3500/500 + 500/30 = 7 + 16.7 = 23.7$ mm; $e_{actual} = M_u/P_u = 150/1.2 = 125$ mm > e_{min}, so design for M_u = 150.
- $P_u / (f_{ck}\, b\, D) = 1200 \times 10^3 / (25 \times 300 \times 500) =$ **0.32**
- $M_u / (f_{ck}\, b\, D^2) = 150 \times 10^6 / (25 \times 300 \times 500^2) =$ **0.08**
- $d'/D = 50/500 =$ **0.10** ⟶ use SP-16 chart for Fe 415, d'/D = 0.1 (steel on two faces, Chart 32).
- From the chart: p/f_{ck} ~ 0.04 (approximately - the exact value is read from your chart) ⟶ $p = 0.04 \times 25 =$ **1.0%**
- $A_{sc} = 1.0 \times 300 \times 500 / 100 =$ **1500 mm^2** ⟶ provide 4 - 22 mm (1520 mm^2).

5. Columns under Biaxial Bending (clause 39.6)

- Corner columns and columns with unsymmetrical beams carry **P_u**, **M_{ux}** **and** **M_{uy}** together. Exact analysis is complex, so IS 456 gives the **Bresler** **load** **contour** method.

Bresler's Formula

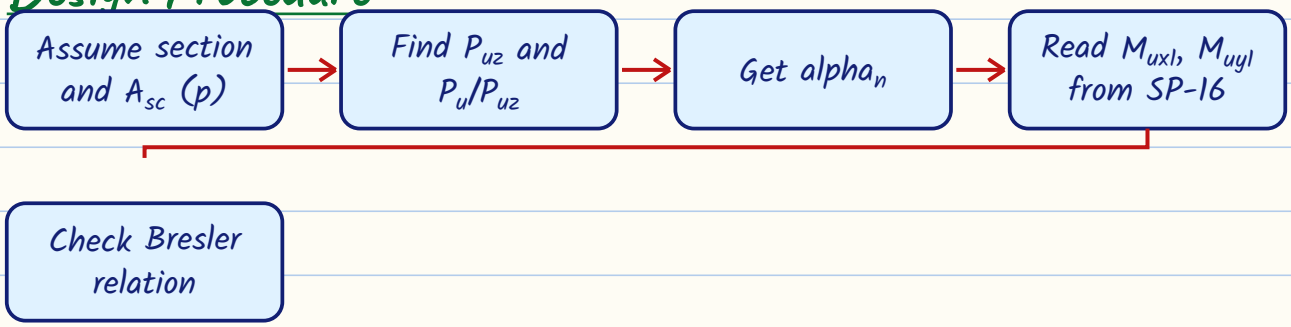
$$(M_{ux} / M_{uxl})^{alpha\ n} + (M_{uy} / M_{uyl})^{alpha\ n} <= 1.0$$

- M_{ux}, M_{uy} = factored moments about x and y axes (including e_{min} effects).
- M_{uxl}, M_{uyl} = **maximum** **uniaxial** **moment** **capacity** about x and y axes for the given P_u (read from SP-16 charts, with steel provided).
- $alpha_n$ depends on P_u/P_{uz}:

$$alpha_n = 1 + (P_u/P_{uz} - 0.2) / 0.6 \text{ for } 0.2 \text{ to } 0.8$$

- **P_{uz}** (squash load) = $0.45\ f_{ck}\ A_c + 0.75\ f_y\ A_{sc}$

Design Procedure



Worked Example 4 - Biaxial

Q. Check a 400 x 400 mm column, M25, Fe 415, with 1.8% steel (2880 mm^2, d'/D = 0.1) for P_u = 1000 kN, M_{ux} = 120 kN.m, M_{uy} = 80 kN.m.

- A_c = 160,000 - 2880 = 157,120 mm^2
- P_{uz} = 0.45 x 25 x 157,120 + 0.75 x 415 x 2880 = 1,767,600 + 896,400 = **2664 kN**
- P_u/P_{uz} = 1000/2664 = 0.375 ⟶ $alpha_n$ = 1 + (0.375 - 0.2)/0.6 = **1.29**
- $P_u/(f_{ck}\ b\ D)$ = 1000 x 10^3 / (25 x 400 x 400) = 0.25; p/f_{ck} = 1.8/25 =

0.072

- From SP-16 chart: $M_{ul}/(f_{ck}\ b\ D^2) \sim 0.119 \longrightarrow M_{uxl} = M_{uyl} \sim 0.119 \times 25 \times 400^3 =$ **190 kN.m** (chart reading, approximate)
- $(120/190)^{1.29} + (80/190)^{1.29} = 0.553 + 0.328 =$ **0.88 < 1.0 ⟶ safe**.

6. Footings - Introduction

- A **footing (foundation)** spreads the load of a column / wall over a large area of soil so that the **safe bearing capacity (SBC)** is not exceeded and settlement is within limits.

Design Principles (clause 34)

- **Plan area** from **service (working) loads** and SBC: A = (W + weight of footing, about 10% of W) / SBC.
- **Depth, reinforcement** from **factored loads** (limit state), with soil pressure $p_u = 1.5 \times p$ (upward, net of footing weight).
- Soil pressure for structural design is **net upward pressure** (self-weight of footing does not cause bending).

Critical Sections

Check	Critical section	Notes
Bending moment	At the **face of column** (wall)	Cantilever projection
One-way (beam) shear	At **distance d** from column face	$tau_v <= tau_c$ (Table 19)
Two-way (punching) shear	At **d/2** from column face all round	$tau_v <= k_s\ tau_c'$, $tau_c' = 0.25\ sqrt(f_{ck})$
Development length	At column face and at the changes of section	$L_d = 0.87\ f_y\ phi / (4\ tau_{bd})$

- $k_s = 0.5 + beta_c <= 1$, where $beta_c$ = short side / long side of the column.

7. Design of Wall Footing

- Main steel is **across** the wall (transverse); **distribution** steel is **along** the wall (0.12% of area).

Procedure

Load per metre W → Width $B = 1.1\,W$ / SBC → $p_u = 1.5\,W / B$ → Projection a

$M_u = p_u\,a^2/2$ → d from shear and moment → A_{st}, distribution steel

Worked Example 5

Q. Design a strip footing for a 230 mm thick wall carrying **150 kN/m** (service) on soil of **SBC 150 kN/m²**. Use M20 and Fe 415.

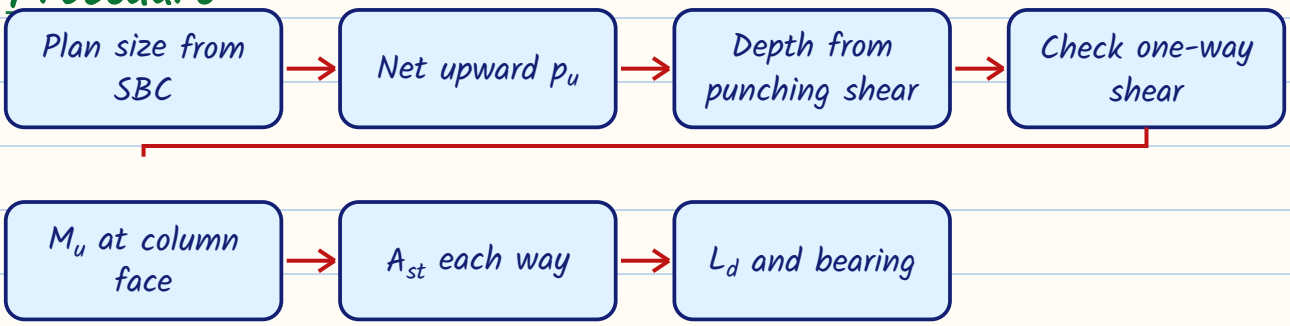
- Width: $B = 1.1 \times 150 / 150 =$ **1.1 m**.
- Net factored pressure: $p_u = 1.5 \times 150 / 1.1 =$ **204.5 kN/m²**.
- Projection $a = (1.1 - 0.23) / 2 =$ **0.435 m**.
- Mu at face of wall $= p_u\,a^2 / 2 = 204.5 \times 0.435^2 / 2 =$ **19.35 kN.m per m**.
- Depth from moment: $d^2 = 19.35 \times 10^6 / (0.138 \times 20 \times 1000) = 7011 \longrightarrow d = 84$ mm (small). **Shear** and min edge depth govern. Try $D = 250$ mm, $d = 250 - 50 - 5 =$ **195 mm**.
- One-way shear at d from face: $V_u = 204.5\,(0.435 - 0.195) = 49.1$ kN; $\tau_v = 49.1 \times 10^3 / (1000 \times 195) =$ **0.25 N/mm²**.
- τ_c for $p_t \sim 0.15\%$ (M20) ~ 0.28 N/mm² > 0.25. **Safe**.
- A_{st}: $19.35 \times 10^6 = 0.87 \times 415 \times A_{st} \times 195\,(1 - \ldots) \longrightarrow A_{st} \sim 289$ mm²; minimum steel $= 0.12\% \times 1000 \times 250 =$ **300 mm²**.
- Provide **8 mm @ 150 mm c/c** (335 mm²) across the wall.
- Development length $L_d = 0.87 \times 415 \times 8 / (4 \times 1.2 \times 1.6) = 376$ mm; available $= 435 - 50 = 385$ mm $>= 376$ **OK**.
- Distribution steel $= 0.12\% \times 1000 \times 250 = 300$ mm² → **8 mm @ 160**

mm c/c along the wall.

> Final: B = 1.1 m, D = 250 mm, 8 mm @ 150 main (transverse), 8 mm @ 160 distribution.

8. Isolated Rectangular (Pad) Footing - Axial Load

Procedure



- **Plan size**: A = 1.1 P / SBC. Choose L/B equal to column D/b (so that projections are equal) - square for square columns.
- **Depth by punching shear** (critical perimeter at d/2): $b_o = 4 (c + d)$ for a square column of side c.
  - $V_u = p_u [B^2 - (c + d)^2]$; $\tau_v = V_u / (b_o d) \le k_s \tau_c'$
- **Bending moment**: $M_u = p_u B a^2 / 2$ where a = projection from the column face. Same in both directions for square footing.

Worked Example 6

Q. Design a square footing for a **400 x 400 mm** column carrying **1000 kN** service load. SBC = **200 kN/m²**. M25, Fe 415.

- Area = 1.1 x 1000 / 200 = 5.5 m^2 ⟶ **2.4 m x 2.4 m** (5.76 m^2).
- p_u = 1.5 x 1000 / 5.76 = **260.4 kN/m²** (net, upward).
- Projection a = (2.4 - 0.4)/2 = **1.0 m**.
- M_u = 260.4 x 2.4 x 1.0^2 / 2 = **312.5 kN.m**
- **Depth by punching shear** (try d = 490 mm): b_o = 4 (400 + 490) = 3560 mm
  - V_u = 260.4 [5.76 - 0.89^2] = 260.4 x 4.968 = 1294 kN

  - tau_v = 1294 x 10^3 / (3560 x 490) = **0.74** **N/mm²**
  - allowable = k_s x 0.25 sqrt(25) = 1 x 1.25 = 1.25 N/mm² > 0.74 **OK**
- Provide **D = 550 mm**, d = 550 - 50 - 10 = **490 mm**.
- **One-way shear** at d from face: V_u = 260.4 x 2.4 x (1.0 - 0.49) = 318.7 kN. tau_v = 318.7 x 10^3 / (2400 x 490) = **0.27** **N/mm²**.
- **Steel**: A_{st} = 312.5 x 10^6 / (0.87 x 415 x 0.95 x 490) = **1859** **mm²** ⟶ p_t = 0.158% ⟶ tau_c (M25) ~ 0.29 > 0.27 **Safe**.
- Minimum steel = 0.12% x 2400 x 550 = 1584 mm² < 1859 OK.
- Provide **12 mm @ 140 mm c/c** both ways (17 bars = 1923 mm²).
- L_d = 0.87 x 415 x 12 / (4 x 1.4 x 1.6) = 484 mm < available (1000 - 50 = 950 mm) **OK**.

Final: 2.4 x 2.4 m footing, D = 550 mm, 12 mm @ 140 c/c in both directions.

9. Eccentrically Loaded Footing

- Occurs when the column carries a **moment** M (or is not at the centroid of the base). Load P at eccentricity **e = M / P** from the centre line.
- The soil pressure is trapezoidal / triangular:

$$p = P/A\ (1 +/- 6e / L)$$

- p_{max} = P/(BL) (1 + 6e/L) and p_{min} = P/(BL) (1 - 6e/L), L = length in the direction of eccentricity.
- **No tension condition**: e <= L/6 (load within the **middle third**). Then p_{min} >= 0.

Procedure

Numerical illustration

- P = 800 kN, e = 0.3 m, footing 2.5 m x 2.0 m (L = 2.5 m). A = 5 m², P/A = 160 kN/m².
- 6e/L = 6 x 0.3 / 2.5 = 0.72 ; e = 0.3 < L/6 = 0.417 ⟶ no tension.

- p_{max} = 160 x 1.72 = **275.2 kN/m²**, p_{min} = 160 x 0.28 = **44.8 kN/m²**.
- Check p_{max} <= SBC (say 300 kN/m²) **OK**. Factored: $p_{u\ max}$ = 1.5 x 275.2 = 412.8 kN/m².

> Eccentric footing: e <= L/6 is the condition for no tension. p_{max} = (P/A)(1 + 6e/L) is the most asked formula.

10. Sloped (Stepped) Footing

- When the projection is large, the thickness needed at the column face is large while at the edge only a small depth is needed. A **sloped (tapered)** footing saves concrete.
- **Minimum edge thickness**: 150 mm (on soil); slope should be gentle (usually <= 35 degrees, so that the top surface can be cast without top formwork).

11. Combined Rectangular Footing for Two Columns

When used

- An **exterior column is on the property line** and cannot project outside.
- Combined footing is a **rectangle** if the two column loads are nearly equal or the outer edge is not restricted, **trapezoidal** if the loads are unequal and the length is restricted; **strap** if the columns are far apart.

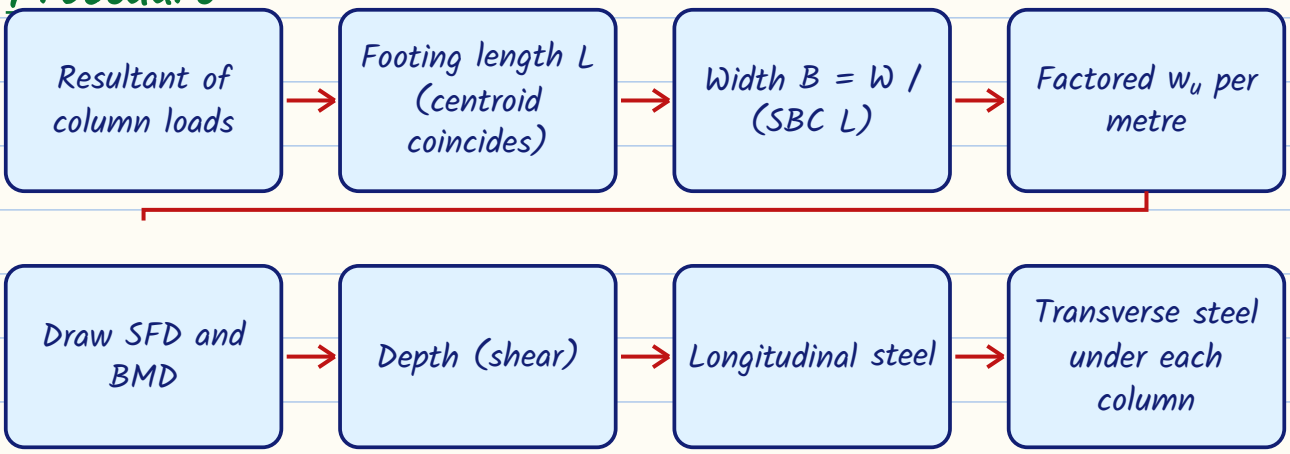
Basic Principle

- The **centroid of the footing area must coincide with the centroid (resultant) of the two column loads**, so that the soil pressure is **uniform**.

$$x_{bar} = P_2\ s\ /\ (P_1 + P_2)\ \text{(from column 1)}$$

- s = centre-to-centre spacing of columns; the footing length L = 2 (x_{bar} + a), where a = distance from the centre of column 1 to the nearer end.
- B = (P_1 + P_2 + 10%) / (SBC x L).

Procedure



- **Longitudinal direction**: footing acts as an **inverted beam** (upward pressure, loads from columns). Maximum **hogging** moment occurs between the columns where the **shear force = 0** ⟶ **top steel**.
- **Transverse direction**: each column acts with a **transverse beam strip** of width (c + d) on each side; the strip acts like a cantilever under the column ⟶ **bottom steel**.
- Check **one-way shear** at d from face of each column and **punching shear** around each column.

Worked Example 7

Q. Design a combined footing for two columns 400 x 400 mm; C1 carries 800 kN and C2 carries 1200 kN (service). Spacing 4 m c/c. SBC = 200 kN/m^2. M25, Fe 415.

- Total service load = 2000 kN; with 10% = 2200 kN. Area = 2200/200 = **11 m^2**.
- Resultant from C1: x_{bar} = 1200 x 4 / 2000 = **2.4 m**.
- Let the left end be 0.4 m from the centre of C1: L = 2 (2.4 + 0.4) = **5.6 m** (right end projects 1.2 m beyond C2).
- B = 11 / 5.6 = 1.96 ⟶ **B = 2.0 m** (area 11.2 m^2).
- Factored loads: P_{u1} = 1200 kN, P_{u2} = 1800 kN, total 3000 kN.
- Upward pressure w_u = 3000 / 5.6 = **535.7 kN/m** (= 267.9 kN/m^2 x 2 m).

Shear force and bending moment (measured from left end)

Section	Shear force (kN)
Left end (x = 0)	0
Just left of C1 (x = 0.4)	535.7 x 0.4 = 214.3
Just right of C1	214.3 - 1200 = -985.7
Just left of C2 (x = 4.4)	535.7 x 4.4 - 1200 = 1157
Just right of C2	1157 - 1800 = -643
Right end (x = 5.6)	-643 + 535.7 x 1.2 = 0

- Zero shear at x = 0.4 + 985.7/535.7 = 0.4 + 1.84 = **2.24 m** from the left end.
- **Maximum hogging moment** M_u = 535.7 x $2.24^2/2$ - 1200 x 1.84 = 1344 - 2208 = **-864 kN.m**.
- **Cantilever moment** at the face of C2 (right side, 1.0 m): 535.7 x $1.0^2/2$ = 268 kN.m (sagging).

Depth - check shear: try D = 1000 mm, d = 940 mm.

- Critical section at d from face of C2 (face at x = 4.2, section at x = 3.26): V_u = 535.7 x 3.26 - 1200 = **546 kN**.
- tau_v = 546 x 10^3 / (2000 x 940) = **0.29 N/mm²**.
- **Longitudinal steel** (top): A_{st} = 864 x 10^6 / (0.87 x 415 x 0.9 x 940) = 2829 mm² ⟶ provide **10 - 20 mm** bars (3142 mm²), p_t = 0.167%, tau_c ~ 0.30 N/mm² > 0.29 **Safe** (shear).
- Minimum steel = 0.12% x 2000 x 1000 = 2400 mm² OK. Bottom steel under the cantilever: provide minimum steel (2400 mm²) ⟶ 16 mm @ 150 mm.
- **Punching shear** at C2: b_o = 4 (400 + 940) = 5360 mm; V_u = 1800 - 267.9 x 1.34^2 = 1319 kN; tau_v = 1319 x 10^3/(5360 x 940) = 0.26 N/mm² < 1.25 **OK**.

Final: L = 5.6 m, B = 2.0 m, D = 1000 mm; top steel 10 - 20 mm longitudinal; bottom min steel + transverse bars under each column.

12. Quick Revision

Formulas to remember

- e_{min} = l/500 + D/30 >= 20 mm
- Axial short column: $P_u = 0.4 f_{ck} A_c + 0.67 f_y A_{sc}$; helical: 1.05 x
- $P_{uz} = 0.45 f_{ck} A_c + 0.75 f_y A_{sc}$; $alpha_n$: 1 (P_u/P_{uz} <= 0.2) to 2 (>= 0.8)
- Bresler: $(M_{ux}/M_{uxl})^{alpha\ n} + (M_{uy}/M_{uyl})^{alpha\ n}$ <= 1
- Eccentric footing: p = (P/A)(1 +/- 6e/L), e <= L/6

Important Questions

- 1. Classify columns. Explain short and long columns with slenderness limits.
- 2. State the code requirements for longitudinal and transverse reinforcement of columns.
- 3. Design a short square / circular column for the given axial load (tied / helical).
- 4. Explain the interaction diagram and design a rectangular column for axial load and uniaxial moment using SP-16.
- 5. Explain the Bresler load contour method for biaxial bending with an example.
- 6. Design a wall footing for a given load per metre and SBC.
- 7. Design an isolated square / rectangular footing for a column carrying axial load.
- 8. Derive the soil pressure equation for an eccentrically loaded footing. What is the condition for no tension?

UNIT - V
Serviceability & Staircases

Design of Reinforced Concrete Structures (23A01502) - handwritten study notes

1. Limit State of Serviceability

Introduction

- A structure must be **safe** (limit state of collapse) and also **useful** during its life (limit state of serviceability, LSS).
- Serviceability = the structure should not **deflect excessively, crack widely, or vibrate** so that appearance, comfort or function is spoiled.
- IS 456:2000 deals with serviceability in **clause 35** (deflection - 23.2, cracking - 35.3, durability - 8).
- Checks are made with **partial safety factor = 1.0** for loads (characteristic / working loads) and for materials.

Collapse check uses factored loads (1.5 x). Serviceability check uses working loads (1.0 x). Never mix the two.

Serviceability Limit States in IS 456

Limit state	Requirement	Clause
Deflection	Final deflection <= span/250	23.2
Deflection (partitions)	After construction <= span/350 or 20 mm	23.2 (b)
Cracking	Crack width <= 0.3 mm (0.2 mm severe)	35.3.2
Durability	Cover, w/c ratio, cement content	8, 26.4
Vibration	Avoid resonance with machines	35.1

- Check is done as **indirect** (span/depth ratios, bar spacing) or **direct** (calculation of deflection and crack width).

2. Deflection of RC Members

Limits as per IS 456 (clause 23.2)

- The **final** deflection (including creep and shrinkage) due to all loads <= **span / 250**.
- The deflection **after erection of partitions and finishes** <= **span / 350 or 20 mm**, whichever is less.

Two ways to satisfy the code: (1) **Indirect** - span / effective depth limits (simple, for design). (2) **Direct** - calculate deflection using Annex C (for checking).

Short Term Deflection

- Uses the elastic formula with an **effective moment of inertia** I_{eff} because the section is partly cracked.

$$E_c = 5000 \sqrt{f_{ck}} \ N/mm^2$$

$$\text{Short-term deflection } a_i = k\, w\, L^4 / (E_c\, I_{eff})$$

- $k = 5/384$ for simply supported beam with UDL; $k = 1/8$ for cantilever with UDL ($wL^4/8EI$).
- Effective moment of inertia (Annex C):

$$I_{eff} = I_r / \left(1.2 - (M_r/M)(z/d)(1 - x/d)(b_w/b)\right), \quad I_r <= I_{eff} <= I_{gr}$$

  - I_r = moment of inertia of the **cracked** section, I_{gr} = gross section.

Long Term Deflection (Creep and Shrinkage)

- Under sustained load, concrete **creeps**; deflection increases with time. Shrinkage curvature adds further deflection.
- **Creep deflection** is obtained using the **effective modulus** of concrete:

$$E_{ce} = E_c / (1 + \theta)$$

- θ = creep coefficient (depends on age at loading).

Age at loading	Creep coefficient theta
7 days	2.2
28 days	1.6
1 year	1.1

- Long term deflection = short term (permanent loads, using E_{ce}) + shrinkage deflection.

$$a_{cs} = k_3 \, psi_{cs} \, L^2$$

  - psi_{cs} = shrinkage curvature = k_4 $epsilon_{cs}$ / D, $epsilon_{cs}$ = **0.0003** (total shrinkage strain if no data).
  - k_3 = 0.125 for simply supported, 0.5 for cantilever. k_4 depends on (p_t - p_c) and p_t.

Final deflection $a = a_{cc} + a_{cs}$ (a_{cc} = creep + short term of permanent load)

- Total deflection is the sum of: **short term (total load)** + **creep (permanent load)** + **shrinkage**.

Example of E_{ce}: M20 concrete, Ec = 5000 sqrt(20) = 22,360 N/mm^2. At 28 days theta = 1.6, so E_{ce} = 22360 / 2.6 = 8,600 N/mm^2 (only 38% of Ec).

3. Deflection Control by Span / Depth Ratio

Basic Values (clause 23.2.1)

Support condition	Basic L / d
Cantilever	7
Simply supported	20
Continuous	26

- Valid for spans up to **10 m**. For spans above 10 m (except cantilevers)

multiply by **10 / span (m)**.

- For two-way slabs, basic values are applied on the **shorter span** (spans up to 3.5 m, f_y <= 415 N/mm^2).

$$(L/d)_{allowable} = (L/d)_{basic} \times k_t \times k_c \times k_f$$

- **k_t** = modification factor for **tension reinforcement** (Fig. 4)
- **k_c** = modification factor for **compression reinforcement** (Fig. 5)
- **k_f** = factor for flanged beams (Fig. 6 and 7) based on b_w / b_f.

Modification Factor for Tension Steel (Fig. 4)

- Depends on **percentage of steel** $p_t = 100\ A_{st} / (b\ d)$ and the **service stress** f_s in steel:

$$f_s = 0.58\ f_y\ (A_{st,req} / A_{st,prov})$$

- For Fe 415 with $A_{st,req} = A_{st,prov}$: $f_s = 0.58 \times 415 =$ **240 N/mm^2**. For Fe 500: $f_s =$ **290 N/mm^2**. For Fe 250: 145.
- **Maximum k_t = 2.0.** More steel (higher p_t) ⟶ smaller k_t. Less service stress ⟶ larger k_t.

Modification Factor for Compression Steel (Fig. 5)

- Depends on percentage of compression steel $p_c = 100\ A_{sc} / (b\ d)$. Max k_c = **1.5**.
- Compression steel reduces long term deflection. Typical: p_c = 0.25% gives about 1.08, p_c = 1.0% about 1.25.

Flanged Beams

- Flanged beams have more stiffness, so allowable L/d is **increased**: use Fig. 6 and 7 with the ratio **b_w / b_f**.
- Rule of thumb: for b_w/b_f = 0.3 the factor is about **0.8** on the basic value for p_t calculation. Interpolate for ratios between 0.3 and 1.0.
- The percentage of steel p_t for flanged beams is based on the **web width b_w**.

Worked Example 1 - Deflection check of a beam

Given: Simply supported RC beam, effective span 6 m, b = 230 mm, d = 450 mm. Provided 4 bars of 20 mm dia ($A_{st,prov}$ = 1256 mm^2), $A_{st,req}$ = 1100 mm^2. Provided 2 bars of 16 mm in compression (A_{sc} = 402 mm^2). Fe 415, M20. **Check deflection.**

- Basic L/d = **20** (simply supported, span < 10 m).
- p_t = 100 x 1256 / (230 x 450) = **1.21 %**
- f_s = 0.58 x 415 x (1100 / 1256) = 240.7 x 0.876 = **211 N/mm^2**
- From Fig. 4: for p_t = 1.21 % and f_s = 211, **k_t = 1.1** (approx.)
- p_c = 100 x 402 / (230 x 450) = 0.39 %. From Fig. 5, **k_c = 1.1**
- Allowable L/d = 20 x 1.1 x 1.1 = **24.2**
- Actual L/d = 6000 / 450 = **13.3** < 24.2 hence **safe in deflection**.

If actual L/d is more than allowable: increase depth d, or increase steel (reduces f_s), or provide compression steel. Redesign and re-check.

Worked Example 2 - Minimum depth from deflection

Given: Cantilever slab with projection 1.5 m, M20, Fe 415, assumed k_t = 1.4 (p_t about 0.3 %).

- Basic L/d for cantilever = 7; allowable L/d = 7 x 1.4 = **9.8**
- d >= 1500 / 9.8 = **153 mm**. Provide d = 155 mm, D = 180 mm (cover 20 + 5).
- This is the **depth for stiffness**, which is checked later for strength (Mu,lim).

4. Cracking in RC Members

Why do cracks form?

- Concrete is weak in tension (about 10% of compressive strength): it cracks when the tensile strain exceeds about **0.0001 to 0.0002**.
- Cracks are **normal** in RC; what we control is **width** (appearance, durability, corrosion of steel).

Causes and Types of Cracks

Type	Cause	Time
Plastic shrinkage	Rapid evaporation of surface water	First hours
Plastic settlement	Settlement of fresh concrete over bars / formwork	1-3 hours
Early thermal	Heat of hydration, cooling	1 day - 2 weeks
Drying shrinkage	Loss of moisture, restrained	Weeks / months
Flexural cracks	Tension due to bending (vertical cracks)	Under load
Shear / diagonal	Principal tension near supports	Under load
Corrosion cracks	Rusting expands steel (7 x volume)	Years
Bond / splitting	Poor anchorage along bars	Under load

Crack Width Limits (IS 456 clause 35.3.2)

- **0.3 mm** at the tension face for members in **normal exposure** (mild, moderate).
- **0.2 mm** for members in **severe / aggressive exposure** (sea water, chemicals) and **0.1 mm** for water-retaining structures (IS 3370).
- Crack width is measured at the level of steel surface nearest to the tension face.

Control of Cracking: Indirect Method

Cracks are controlled by detailing rules (clause 26.3.3) without calculating crack width:

- **Maximum spacing** of tension bars: **clear distance** between bars should not exceed the values in **Table 15** (depend on f_y and moment redistribution).

f_y (N/mm^2)	Max clear distance, 0% redistribution
250	300 mm
415	180 mm
500	150 mm

- Distance reduces for **negative redistribution** (e.g. -30%: 215 / 125 / 105 mm for Fe 250 / 415 / 500).
- **Side face reinforcement**: if depth of web **D > 750 mm**, provide 0.1% of web area (b_w x D) distributed equally on both faces, spacing <= **300 mm or web thickness, whichever is less**.
- **Minimum bar spacing**: greater of bar diameter and (maximum size of aggregate + 5 mm). Minimum cover as per exposure.

Rule: use **many small bars** instead of few big bars: better crack distribution and bond.

Direct Method: Crack Width Calculation (Annex F)

$$w_{cr} = 3\, a_{cr}\, \text{epsilon}_m / (1 + 2\,(a_{cr} - c_{min}) / (h - x))$$

- a_{cr} = distance from the point considered to the surface of the nearest longitudinal bar.
- c_{min} = minimum clear cover to the longitudinal bar.
- **h** = overall depth of the member, **x** = depth of neutral axis (cracked section).
- **epsilon_m** = average strain at the level considered, obtained after deducting **tension stiffening** of concrete from the steel strain.

Measures to Control Cracking

- Provide **adequate cover** (but not excessive) and use closely spaced small bars.
- Reduce **steel stress** (use more steel or bigger section).

Serviceability summary: Deflection by L/d x modification factors. Cracking by bar spacing (Table 15), side face steel for D > 750 mm, w_{cr} <= 0.3 mm.

5. Staircases: Introduction

- A **staircase** is a sequence of steps that provides access between floors. A stair slab is designed like a **one way slab** (inclined).

Terms used in Stairs

- **Tread (T)**: horizontal part of a step on which foot rests. **Going**: horizontal distance covered by the whole flight (no. of treads x tread).
- **Rise (R)**: vertical height of a step. **Riser**: vertical face of the step.
- **Waist**: thickness of the slab (measured perpendicular to the slope).
- **Flight**: a series of steps between two landings. **Landing**: platform at the end of a flight, for rest and change of direction.

Proportions of Steps

- **Rise R = 150 to 190 mm** (residential 150-175; public buildings 150). **Tread T = 250 to 300 mm**.
- Thumb rule: **2R + T = 550 to 700 mm** (or R + T = about 400 mm).
- Width of stair: residential **1.0 to 1.25 m**; public **1.5 to 2.0 m**; maximum 12-15 risers in one flight.
- Slope of stair: 25 to 40 degrees (normal 30 to 35 degrees).

Types of Staircases

- **Straight stair**: single flight / several flights in one line. Simple; needs long space.
- **Quarter-turn stair**: 90 degrees turn through a landing (L shaped).
- **Half-turn / Dog-legged stair**: two flights in opposite directions with a **mid landing**; **no well**, most common, compact.
- **Open well stair**: two or more flights around a **well (opening)** between them; used in public buildings.
- **Geometrical stair**: open well with **curved** turn, no newel (the handrail is a smooth curve).
- **Spiral stair**: steps radiate from a central post / column; takes least floor space, used for fire escape or limited access.
- **Helicoidal stair**: slab twisted in helix shape without central support;

torsion governs; costly.

> Dog-legged stairs are most popular for residential buildings and design problems.

6. Loads on Staircases

Dead Load

- **Self weight of waist slab on slope** = 25 x D (kN/m² of slope). Convert to horizontal plan area:

$$w_s = 25\ D \times \sqrt{R^2 + T^2} / T \ (kN/m^2 \text{ of plan})$$

- **Weight of steps** (triangular): $w_{step} = 25 \times R / 2$ (kN/m² on going).
- **Finishes**: 0.6 to 1.0 kN/m² (tiles, plaster). Landing: 25 x D + finishes.

Live Load (IS 875 Part 2)

Building	LL on stairs
Residential, offices (not crowded)	3.0 kN/m²
Public buildings, assembly halls	5.0 kN/m²
Where stairs may be overcrowded	5.0 kN/m²

- Live load acts on the **horizontal projection**. Handrail load 0.75 kN/m as per IS 875.
- **Design load** $w_u = 1.5\ (DL + LL)$.

Effective Span of Stairs (clause 33.1)

- **Stairs spanning longitudinally** (supported at the ends of flight): effective span = c/c of supports, or clear span + effective depth, **whichever is less**.
- If the **landing slab spans in the same direction** as the flight: effective span = **going + the width of the landing** supports (c/c).
- If the **landing spans at right angles** to the flight: effective span = going + at each end **half width of landing or 1.0 m**, whichever is

smaller.

Staircase Spanning Longitudinally vs Transversely

Longitudinal (along the flight)	Transverse (across the width)
Slab supported at bottom and top landing beams	Supported on side walls or stringer beams
Span = going + landings	Span = width of stair (small)
Steps carried by waist slab	Steps act as beams / waist is thin
Used for dog-legged, common	Used for cantilever steps from a wall, open stairs
Main steel along slope	Main steel along width

Whenever possible provide the support to the landing on **both** sides; this lowers the span, cost and deflection.

7. Design of a Dog-Legged Staircase

Design Steps

Fix R, T, no. of steps → Effective span L → Assume depth (L/d x MF) → Calculate loads (going, landing)

BM and SF

Check d from Mu,lim → Find A_{st} → Distribution steel → Check shear, deflection

Check development length, detailing

Points to Remember

- Slab is designed per metre **width** of the stair. Loads are on **horizontal plan** area.
- Bending moment (simply supported): $M = w L^2 / 8$. When the slab is continuous with the landing/beams, take $w L^2 / 10$ (as per IS 456 Annex 22.5, end spans) for midspan. For stairs built into walls, $w L^2 / 10$ is also used.
- For stairs spanning with landing in the same direction: calculate loads separately on flight and landing, find reactions and **maximum BM** at the section of zero shear.
- Depth of waist slab: $L / 20$ **to** $L / 25$ for simply supported; $L / 26$ **to** $L / 30$ with continuity.
- Main steel along the slope. **Distribution steel**: 0.12% of gross area (HYSD), perpendicular to main bars.
- At the junction of flight and landing in dog-legged stairs, **provide extra bars** (re-entrant corner bars) and tie so that the bars do not straighten and burst the concrete.

Worked Example 3: Design of a Dog-Legged Staircase

Given: Floor to floor height **3.2 m**. Two equal flights, stair width **1.2 m**, landing width 1.2 m. Live load **3 kN/m²**, floor finish 1.0 kN/m². **M20 and Fe 415.** Landing slab spans in the same direction, resting on 230 mm walls. Design the flight.

Step 1: Proportions

- Height of one flight = 3.2 / 2 = **1.6 m**. Rise R = 160 mm, so number of risers = 1600 / 160 = **10**.
- Treads per flight = 10 - 1 = **9**; tread T = **250 mm**; going = 9 x 0.25 = **2.25 m**.

Step 2: Effective span

- Flight (going) = 2.25 m, plus landing 1.2 m in the same direction.
- Effective span L = 2.25 + 1.2 = **3.45 m** (c/c of supports, taking the

walls as simple supports).

Step 3: Trial depth

- Take L/d = 20 x 1.5 (modification) = 30, so d = 3450 / 30 = 115 mm. **Take D = 170 mm, d = 145 mm** (cover 20 mm + bar 12/2 = 5 mm; D = 145 + 25).

Step 4: Loads (per m width)

Item	Calculation	kN/m²
Waist slab on slope	25 x 0.17 x sqrt(0.16²+0.25²)/0.25 = 4.25 x 1.187	5.05
Steps	25 x 0.16 / 2	2.00
Finishes	given	1.00
Live load	given	3.00
Total on going w_1	DL + LL	11.05
Landing: self weight	25 x 0.17	4.25
Landing: finish + LL	1.0 + 3.0	4.00
Total on landing w_2		8.25

- Factored loads: going **w_{u1} = 1.5 x 11.05 = 16.58 kN/m**; landing **w_{u2} = 1.5 x 8.25 = 12.38 kN/m**.

Step 5: Bending moment and shear

- Load on going = 16.58 x 2.25 = 37.30 kN (acts at 1.125 m from A); load on landing = 12.38 x 1.2 = 14.86 kN (acts at 2.85 m from A).
- Taking moments about A: R_B = (37.30 x 1.125 + 14.86 x 2.85) / 3.45 = **24.44 kN**; R_A = 52.16 - 24.44 = **27.72 kN**.
- Zero shear at x = 27.72 / 16.58 = **1.672 m** from A (lies in the going).
- **M_u = 27.72 x 1.672 - 16.58 x 1.672² / 2 = 46.35 - 23.18 = 23.17 kN.m per m**.
- Check with w L^2 / 8: average w = 52.16/3.45 = 15.12; M = 15.12 x 3.45² / 8 = 22.5 kN.m (close).
- Design shear V_u = **27.72 kN**.

Step 6: Check depth

- $M_{u,lim} = 0.138\ f_{ck}\ b\ d^2 = 0.138 \times 20 \times 1000 \times 145^2 =$ **58.0 kN.m** > 23.17 kN.m, so section is under-reinforced. **d is OK**.

Step 7: Main steel

$$M_u / (b\ d^2) = 23.17 \times 10^6 / (1000 \times 145^2) = 1.10\ N/mm^2$$

$$A_{st} = 0.5\ (f_{ck}/f_y)\ [\ 1 - sqrt(1 - 4.6\ M_u/(f_{ck}\ b\ d^2))\ ]\ b\ d = 476\ mm^2$$

- Minimum steel = 0.12% x 1000 x 170 = 204 mm^2 < 476 mm^2, OK.
- Spacing of 12 mm bars = 1000 x 113 / 476 = 237 mm. **Provide 12 mm dia @ 200 mm c/c** ($A_{st,prov}$ = 565 mm^2); max spacing 3d = 435 or 300 mm, OK.

Step 8: Distribution steel

- A_{st} = 0.12% x 1000 x 170 = **204 mm^2**. Using 8 mm bars: spacing = 1000 x 50.3 / 204 = 246 mm. **Provide 8 mm @ 240 mm c/c** (max Sd = 725 or 450 mm).

Step 9: Check for shear

- $tau_v = V_u / (b\ d)$ = 27720 / (1000 x 145) = **0.19 N/mm^2**.
- p_t = 100 x 565 / (1000 x 145) = 0.39 %, so tau_c = 0.43 N/mm^2 (Table 19, M20). For slab thickness 170 mm, k = 1.26, so k tau_c = **0.54** > 0.19. **Safe in shear** (no shear steel).

Step 10: Check for deflection

- Basic L/d = 20. f_s = 0.58 x 415 x (476/565) = **203 N/mm^2**. From Fig. 4 with p_t = 0.39: k_t = **1.5**.
- Allowable L/d = 20 x 1.5 = **30**. Actual L/d = 3450 / 145 = **23.8** < 30. **Safe in deflection.**

Step 11: Development length

- $L_d = 0.87\ f_y\ phi / (4\ tau_{bd})$; tau_{bd} = 1.2 x 1.6 = 1.92 N/mm^2 (deformed bars). L_d = 0.87 x 415 x 12 / (4 x 1.92) = **564 mm (47 phi)**. Bars must be extended at least this length into supports.

Detailing

- Main bars: 12 mm @ 200 mm c/c along the slope, bent up at the landing. Provide **top steel (0.5 of bottom)** at supports for continuity: 10 mm @ 250 mm c/c for length L/5 to L/4.
- Distribution bars: 8 mm @ 240 mm c/c on top of main bars. Clear cover 20 mm.
- At the junction of the two flights, the bars should be bent properly and **cross each other**, not be continuous in a straight line.

Key point: Check list for stairs - rise/tread, effective span, loads on slope converted to plan, $M = wL^2/8$ or max BM by reactions, d from Mu,lim, Ast, distribution steel (0.12%), shear ($tau_v < k\ tau_c$), deflection (L/d), Ld.

8. Revision Notes

- **Serviceability** = deflection + cracking + durability + vibration. Checks use **working loads**.
- Deflection limits: **L/250 total**; **L/350 or 20 mm** after partitions.
- Basic **L/d** = 7 (cantilever), 20 (simply supported), 26 (continuous). Multiply by k_t, k_c, k_f. Reduce by 10/span for span > 10 m.
- **$f_s = 0.58\ f_y\ (A_{st,req}/A_{st,prov})$**. $E_{ce} = Ec / (1 + theta)$. Ec = 5000 sqrt($f_{ck}$).
- Cracking: **0.3 mm**, Table 15 bar spacing, side face steel if D > 750 mm (0.1% of web area), Annex F crack width formula.

Important Questions

- 1. What are the limit states of serviceability? Explain the serviceability requirements of IS 456.
- 2. How is deflection controlled in RC beams? Explain the modification factors for tension and compression steel.
- 3. Explain short term and long term deflections. What is the effect of

creep and shrinkage?

- 4. A simply supported beam of span 6 m has b = 230, d = 450, 4 - 20 mm bars. Check the deflection.
- 5. What are the causes of cracks in RC members? Explain the methods to control cracks.
- 6. State the clause 26.3.3 rules for the spacing of reinforcement. What is the side face reinforcement?
- 7. Explain the different types of staircases with neat sketches.
- 8. Design a dog-legged staircase for a floor height of 3.2 m. Live load 3 kN/m^2, M20, Fe 415. Draw the reinforcement details.